

Role of tankyrase scaffolding in the β -catenin destruction complex and WNT signaling

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eLife Assessment

This **important** study reports the development of the first tankyrase degrader and demonstrates its enhanced ability to inhibit β -catenin signaling compared to conventional tankyrase inhibitors. The evidence supporting the conclusions is comprehensive and **convincing**, based on rigorous biochemical and cellular analyses. The findings will be of broad interest to researchers studying Wnt signaling, protein degradation, and cancer biology.

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Abstract

Aberrant WNT/ β -catenin signaling drives tumorigenesis and metastasis in cancer. Although enzymatic inhibitors of tankyrase (TNKS) effectively block AXIN degradation and stabilize the β -catenin destruction complex (DC), they have demonstrated limited efficacy in various cancer models. Here we demonstrate that, unexpectedly, the induction of AXIN puncta represents a major barrier to achieving therapeutic efficacy. Mechanistically, catalytic inhibition of TNKS prevents TNKS turnover and drives its accumulation in the DC, wherein the scaffolding function of TNKS induces AXIN puncta formation, rigidifies the DC, and impedes β -catenin turnover. Chemically induced degradation of TNKS overcomes this limitation by stabilizing AXIN without puncta formation, providing a deeper suppression of the WNT/ β -catenin pathway activity and the proliferation of colorectal cancer cells harboring dysfunctional APC mutations. Collectively, these findings provide an explanation for the unsatisfactory outcomes of drugging the WNT/ β -catenin signaling pathway by TNKS inhibitors and highlight TNKS degradation as a promising approach to treat WNT/ β -catenin-driven cancers.

Introduction

WNT/ β -catenin signaling plays an important role in cell growth, differentiation, and migration^{1,2}. Dysregulation of this pathway enables reprogrammed cancer cells to proliferate, metastasize and resist chemo- and radiotherapies. In colorectal cancer (CRC), loss-of-function mutations in adenomatous polyposis coli (APC) are the major drivers of aberrant WNT/ β -catenin signaling³. However, therapeutic targeting this oncogenic pathway remains unsuccessful despite extensive efforts⁴. For example, although silencing tankyrase (TNKS1/2, also known as PARP5A/B and ATRD5/6)^{5,6-7} phenocopied APC restoration and prevented tumorigenesis in APC-null mice⁸, TNKS inhibitors (TNKSi) failed to show promising efficacy in various in vitro and in vivo CRC models. Addressing the discrepancy between the genetic and pharmacological interception of WNT/ β -catenin signaling through TNKS may provide a path toward developing new CRC treatments.

The axis inhibition protein (AXIN) and APC are both scaffolding proteins of the β -catenin destruction complex (DC) that exists as biomolecular condensates in the cytoplasm to prime β -catenin for proteasomal degradation⁹⁻¹¹. As TNKS-catalyzed poly(ADP-ribose)-dependent ubiquitination (PARdU) is required for the turnover of AXIN¹²⁻¹⁴, blocking the enzymatic function of TNKS by IWR1 or XAV939 leads to AXIN accumulation^{12,15}, thereby enhancing the ability of the DC to promote β -catenin degradation. The accompanied AXIN puncta formation is generally attributed to AXIN accumulation and is considered to play a positive role in suppressing WNT/ β -catenin signaling. However, TNKS also associates with the DC^{16,17}, where its contributions to the DC function remains poorly understood. Specifically, because TNKS regulates its own abundance through self-poly(ADP-ribosylation) (PARylation), catalytic inhibition also promotes drastic TNKS accumulation¹². It is unclear whether this feedback mechanism affects the DC function and results in the limited anticancer efficacy of TNKSi, as inhibitor-induced TNKS accumulation may support WNT/ β -catenin signaling through molecular scaffolding independently of its catalytic function. Indeed, deletion of the catalytic PARP domain of TNKS attenuated, but did not fully suppress the WNT-induced transcription of the TCF/LEF-controlled genes^{18,19}. By contrast, removal of the sterile alpha motif (SAM) domain or the ankyrin repeat clusters (ARCs) of TNKS abolished the WNT/ β -catenin pathway activity more completely.

To understand the role of TNKS scaffolding in WNT/ β -catenin signaling, we developed a TNKS-targeting proteolysis-targeting chimera (PROTAC)^{20,21} as a selective TNKS degrader (TNKSd). Comparison between TNKSd and TNKSi allowed us to understand the catalysis-independent scaffolding function of TNKS and uncover the mechanism underlying the ineffectiveness of TNKSi against cancers. We found that inhibiting TNKS by TNKSi led to the accumulation of TNKS within the DC, which induced AXIN puncta formation, promoted maturation^{22,23} of the DC condensates, and impeded β -catenin turnover. In contrast, degrading TNKS by TNKSd enriched AXIN without inducing puncta or impairing the dynamics of the DC. As a result, TNKSd provided deeper suppression of the WNT/ β -catenin pathway activity as well as better control of the proliferation of CRC cells harboring dysfunctional APC mutations. Collectively, our study provides new insights into the roles of TNKS in the DC and presents a promising strategy to enhance the therapeutic efficacy of TNKS-targeting anticancer treatments.

Results

IWR1-POMA induces TNKS degradation

To study the role of TNKS scaffolding, we first developed a PROTAC molecule to enable selective degradation of TNKS1 and TNKS2 at the same time by small molecules. Compared to conventional genetic approaches, this chemical approach helps address gene redundancy more conveniently and effectively. TNKS uses nicotinamide adenine dinucleotide (NAD⁺) as the ADP-ribose donor. The corresponding pocket in the PARP domain contains two discrete sites, one for nicotinamide (NI) and the other for adenosine (AD) binding. IWR1 is a highly selective TNKSi that exploits the AD site unique to TNKS, and IWR6 is a promiscuous but potent TNKSi structurally similar to XAV939 that binds to the NI site common to all PARPs^{12,24}. Based on the crystal structures of TNKS1 and TNKS2 in complex with IWR1 or XAV939 (**Figure S1**), we envisioned that modifying the quinoline group of IWR1 and the phenyl group of IWR6 would not negatively impact TNKS-binding. We have thus created a small library of 30 PROTAC molecules by attaching pomalidomide²⁵ or VH032²⁶ through a PEG chain of various lengths to IWR1 or IWR6 (**Figure S2A**) for engaging cereblon (CRBN) or the von Hippel-Lindau tumor suppressor protein (VHL), respectively, and examined their ability to induce TNKS degradation.

To facilitate the discovery of a TNKSd, we also developed a luciferase assay to measure the level of TNKS1 in a high-throughput manner. Using the CRISPR-assisted insertion tagging (CRISPaint) technology²⁷, we introduced a nanoluciferase (NanoLuc) tag to the C-terminus of TNKS1 in HAP1 cells (**Figure S2B**). We chose to monitor TNKS1 because it is constitutively expressed while TNKS2 is expressed minimally without induction. By following the luciferase activity, the knockdown efficiency of all 5 series of the PROTAC molecules could be quantified easily and accurately using a robotic liquid dispenser system in 96-well plates, which significantly improved the efficiency over the traditional method that relies on Western blotting and image processing. Using this luciferase assay, we found that the IWR1-based PROTACs eliminated TNKS1 significantly more effectively than the IWR6-based PROTACs (**Figure S2C**). Additionally, CRBN mediated TNKS1 degradation more efficiently than VHL. Inserting a triazole group between the quinoline group and the PEG chain further enhanced the degradation efficiency. However, replacing the PEG linker with a nonpolar or rigid linker did not improve the efficacy. Meanwhile, employing G007-LK²⁸ as the warhead with various linker strategies led to reduced potency and degree of degradation.

IWR1-TP4-Poma (IWR1-POMA hereafter) is the most potent PROTAC among these series. It initiated TNKS1 degradation at 1 nM, operated with a DC₅₀ value of 60 nM, and reached 96% degradation at 1.2 μM in HAP1 cells (**Figure 1A**). Western blot analysis confirmed that IWR1-POMA could also deplete TNKS1 and suppress the induction of TNKS2 in a dose and time-dependent manner in DLD-1 CRC cells that carry truncating APC mutations (**Figure 1B and C**). The level of TNKS did not recover 36 hours after IWR1-POMA was removed, indicating a durable drug effect (**Figure S3A**). Addition of IWR1, pomalidomide or MG132 blocked TNKS degradation (**Figure S3B** and **C**), supporting that IWR1-POMA functioned through the designed mode of action. The generality of IWR1-POMA to induce TNKS degradation was demonstrated in SW480, HT-29 and HeLa cells (**Figure S3D**). The ability of IWR1-POMA to stabilize AXIN and promote β-catenin degradation in the cytoplasm was confirmed further in HEK293 cells (**Figure 1D**). Compared to IWR1, IWR1-POMA provided better control of the cytosolic β-catenin level. The performance of this PROTAC molecule is particularly notable, considering that shutting down the catalysis of TNKS leads to a >30-fold accumulation of total TNKS, creating a nearly 3-orders-of-magnitude difference in the levels of TNKS between catalytic inhibition and chemically induced degradation. To our knowledge, this is the first example of targeted degradation of an auto-regulated protein.

To examine the degradation specificity of IWR1-POMA, we performed proteome-wide expression profiling on DLD-1 cells treated with DMSO, IWR1 or IWR1-POMA for 16 h. Using a tandem mass tag (TMT)-based, multiplexed quantitative mass spectrometric (MS) approach with two biologically independent replicate samples (**Figure S4A**), we identified 7,884 proteins, among which 5,822

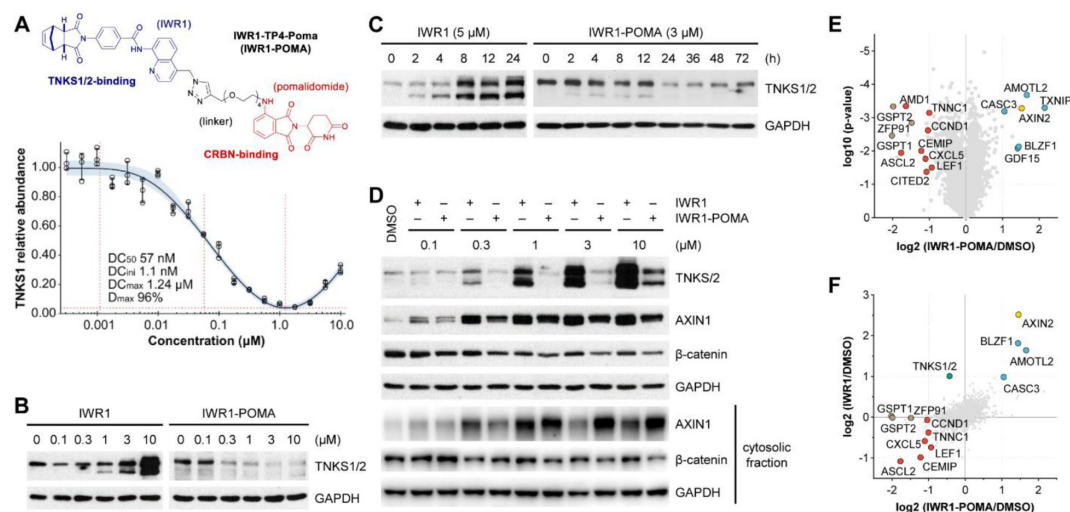


Figure 1.

Characterization of IWR1-POMA

(A) IWR1-POMA induced TNKS degradation with a DC50 value of 60 nM and reached a nearly complete depletion of TNKS1 at 1.2 μ M in HAP1 cells. The dose-response curve was presented with 95% confidence interval (CI). (B and C) IWR1 induced massive TNKS accumulation while IWR1-POMA promoted deep degradation in a dose and time-dependent manner in DLD-1 cells. (D) IWR1-POMA promoted a more complete degradation of β -catenin than IWR1 in HEK293 cells cultured with Wnt3A conditioned media. (E and F) TNKS substrates (light blue) accumulated and WNT/ β -catenin-controlled proteins (red) down-regulated in DLD-1 cells treated with IWR1-POMA (3 μ M). AXIN2 (yellow) is both a TNKS substrate and a WNT target. GSPT1/2 and ZFP91 (light brown) are the only off-targets identified. Comparative analysis showed that IWR1-POMA controlled WNT/ β -catenin signaling more effectively than IWR1.

could be quantified with high confidence (<1% FDR). Correlation analysis confirmed the consistency of the quantitative analysis of the two biological replicates (**Figure S4B**). However, accurate quantification of TNKS1 proved challenging, as only a low-abundance peptide corresponding to TNKS1 was detected in this experiment. Nonetheless, IWR1-POMA induced the accumulation of AXIN2 along with other TNKS substrates (**Figure 1E**). LEF1 and several other WNT targets were also effectively downregulated, confirming the on-target drug effects. Pairwise binary comparison of the drug treatment versus control indicated better control of WNT signaling by IWR1-POMA in addition to high degradation specificity (**Figure 1F**). GSPT1/2 and ZFP91, three common off-targets of the IMiD-based PROTACs, were the only perturbations not related to the WNT pathway. None of the other 7 PARPs and 81 NAD⁺/NADP⁺-dependent enzymes identified in this MS experiment were significantly affected by IWR1-POMA (**Figure S4C**).

PROTAC suppresses catalysis-independent WNT/ β -catenin signaling by TNKS

A previous study showed that polymerization of PARylation-incompetent TNKS can drive WNT/ β -catenin signaling¹⁸. To understand whether the lack of efficacy of TNKSi in suppressing CRC cell growth originated from the PARylation-independent WNT/ β -catenin signaling induced by the accumulated TNKS upon catalytic inhibition, we treated Wnt3A-stimulated 293T cells with IWR1 or IWR1-POMA. We reason that, by removing both the catalysis-dependent and independent functions of TNKS, IWR1-POMA may suppress WNT/ β -catenin signaling more completely than IWR1 that inhibits AXIN PARylation but also promotes TNKS accumulation and polymerization. Indeed, IWR1-POMA promoted β -catenin degradation significantly more effectively than IWR1 under various doses of Wnt3A (**Figure S5A**). To investigate the individual contribution of TNKS1 and TNKS2 to WNT/ β -catenin signaling, we co-transfected a low dose of TNKS1 plasmid together with SuperTopFlash (STF), a luciferase reporter for the WNT/ β -catenin activity, into 293T-TNKS1/2-DKO cells wherein both TNKS1 and TNKS2 were deleted by CRISPR²⁹ (**Figure S5B**). We found that addition of IWR1 suppressed the TNKS1-induced STF activity but not completely (**Figure 2A** and **S5C**). There was a small but noticeable residual activity, which could not be removed by increasing the concentration of IWR1. In contrast, IWR1-POMA induced TNKS1 degradation and promoted deeper suppression of the STF activity. Similarly, IWR1-POMA reduced the level of TNKS2 and provided better control of the STF activity than IWR1 in TNKS1/2-DKO cells expressing TNKS2 (**Figure 2B** and **S5D**). As such, both TNKS1 and TNKS2 can contribute to WNT signaling that persists even in the presence of catalytic inhibition of TNKS PARylation.

We next used PARylation-incompetent variants of TNKS to verify whether TNKS1 and TNKS2 can promote WNT/ β -catenin signaling independently of their catalytic function. The catalytic site of TNKS is highly conserved. Deactivation of the H-Y-E triad of TNKS1 with H1184A and E1291A mutations³⁰ abolishes the binding of NAD⁺ and the accepting ADP ribose. This PARP-dead (PD) variant induced notable WNT/ β -catenin signaling in the TNKS1/2-DKO cells even at low doses, confirming that TNKS1 has a catalysis-independent functional role in the DC (**Figure 2C** and **S5E**). Addition of IWR1-POMA degraded TNKS1-PD and reduced the pathway activity, whereas treatment with IWR1 had no effect. Similarly, TNKS2-M1054V having an impaired ability to interact with the accepting ADP ribose³¹ induced robust WNT/ β -catenin signaling and was insensitive to IWR1 treatment (**Figure 2D** and **S5F**). In contrast, IWR1-POMA degraded this catalytically inactive variant of TNKS2 and suppressed the corresponding STF activity in a dose-dependent manner. Collectively, these data support the hypothesis that both TNKS1 and TNKS2 play an additional role in the DC to regulate WNT/ β -catenin signaling through a mechanism beyond catalytic PARylation of AXIN¹⁸. Degradation of TNKS eliminates both the enzymatic and non-enzymatic activities of TNKS, providing better control of the WNT/ β -catenin signaling than catalytic inhibition.

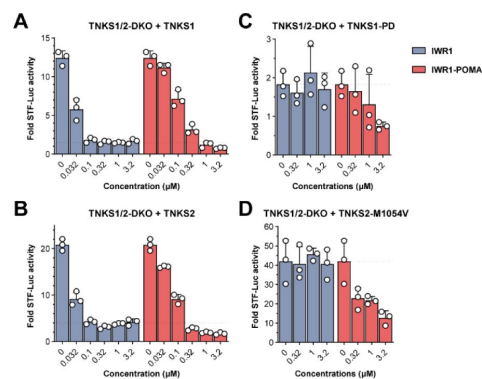


Figure 2.

IWR1-POMA degraded TNKS to suppress both catalysis-dependent and independent WNT signaling

(A) 293T-TNKS1/2-DKO cells were transfected with FLAG-TNKS1 and STF plasmids and then treated with IWR1 or IWR1-POMA. In contrast to IWR1 that plateaued in suppressing the WNT/ β -catenin signaling induced by TNKS1, IWR1-POMA allowed for a more complete control of the pathway activity. (B) 293T-TNKS1/2-DKO cells transfected with FLAG-TNKS2 and STF plasmids and then treated with IWR1 or IWR1-POMA similarly showed that catalytic inhibition led to residual WNT/ β -catenin activity. (C) 293T-TNKS1/2-DKO cells were transfected with 3 $\times$ FLAG-TNKS1-PD and STF plasmids and then treated with IWR1 or IWR1-POMA. IWR1-POMA suppressed the luciferase activities whereas IWR1 had no effect. (D) 293T-TNKS1/2-DKO cells transfected with FLAG-TNKS2-M1054V and STF plasmids and then treated with IWR1 or IWR1-POMA. TNKS2 exhibited robust scaffolding effects. All the data is presented as mean $\pm$ SEM, n = 3.

Tankyrase controls the dynamic assembly of the DC

AXIN interacts with APC to work as a molecular scaffold for the DC to catalyze β -catenin phosphorylation and ubiquitination. TNKS also associates with this complex^{16,17} to control the AXIN level. Although much is known about the catalytic function of TNKS, its scaffolding function^{18,19} is poorly understood. To study the non-enzymatic function of TNKS in WNT signaling, we transfected 293T cells with a low dose of AXIN1-mCherry plasmid and confirmed that TNKS colocalized with AXIN1 to form micrometer-sized puncta (**Figure 3A**). In contrast, in TNKS1/2-DKO cells that lack TNKS, AXIN1-mCherry distributed diffusely throughout the cytoplasm (**Figure 3B**). However, AXIN puncta formed readily when AXIN1-mCherry was expressed together with TNKS1 in the TNKS1/2-DKO cells (**Figure 3C**). Similarly, TNKS2 also induced AXIN puncta in TNKS1/2-DKO cells (**Figure 3D**), suggesting that TNKS1 and TNKS2 play a redundant structural role in the DC to support AXIN puncta formation. To further verify whether AXIN PARylation is involved in the puncta formation, we expressed AXIN1-mCherry together with catalytically inactive variants of TNKS and found that both TNKS1-PD and TNKS2-M1054V could induce AXIN puncta effectively (**Figure 3E and F**). Thus, TNKS1 and TNKS2 promoted AXIN puncta formation through molecular scaffolding independently of their ability to catalyze protein PARylation.

Both IWR1 and IWR1-POMA stabilize AXIN to suppress WNT/ β -catenin signaling. However, IWR1 induces TNKS accumulation while IWR1-POMA promotes TNKS degradation. Given the observation that AXIN puncta formation requires TNKS, we next examined how these TNKS modulators affect the assembly of the DC. As expected, treating SW480 cells with IWR1 induced AXIN puncta while IWR1-POMA did not (**Figure 4A**). Similarly, IWR1 induced AXIN puncta in 293T cells wherein the endogenous AXIN1 was labeled with RFP at its C-terminus by CRISPR³², but IWR1-POMA failed to induce AXIN puncta (**Figure 4B**). As 293T cells harbor wild-type APC and SW480 cells carry a truncating APC without the AXIN-binding sites^{33,34}, the ability of TNKSi to induce AXIN puncta is independent of AXIN-APC binding. These results also suggest that tagging AXIN1 with a fluorescent protein does not affect its ability to oligomerize.

To further understand the functional significance of AXIN puncta, we transfected HeLa cells with a low dose of AXIN1-GFP plasmid. Again, IWR1 promoted the formation of micrometer-sized AXIN puncta that are significantly larger than those observed in the DMSO control sample (**Figure 4C** and **S6A**). In contrast, AXIN1 distributed diffusely throughout the cytoplasm when the cells were treated with IWR1-POMA. To determine the effects of AXIN puncta on WNT signaling, we expressed AXIN1-GFP together with STF in HeLa cells under the same conditions and found that IWR1-POMA suppressed the luciferase activity better than IWR1 (**Figure S6B**). Thus, AXIN puncta formation is not a prerequisite for the DC to catalyze β -catenin degradation as commonly believed. Instead, the large AXIN puncta induced by IWR1 are less functional DCs than the unaggregated AXIN complexes formed upon IWR1-POMA treatment. We have further confirmed that the N-terminally tagged AXIN1 and the C-terminally tagged AXIN2 also responded to IWR1 and IWR1-POMA treatments in the same manner (**Figure S6C** and **D**).

The DC is a classic example of biomolecular condensates^{9,10}. TNKS interact with both AXIN and APC and can self-aggregate to form insoluble filaments^{16,19,31,35,36}. We suspected that the massive accumulation of TNKS induced by IWR1 rigidified the DC^{22,23} and limited its ability to turn over β -catenin. To test this hypothesis, we expressed AXIN1-mCherry and mNeonGreen- β -catenin in HeLa cells and then treated these cells with DMSO, IWR1 or IWR1-POMA. AXIN1-mCherry and mNeonGreen- β -catenin colocalized in all samples, indicating that labeling AXIN1 and β -catenin with a fluorescent protein did not disrupt the recruitment of β -catenin to the DC (**Figure 5A**). We then performed fluorescence recovery after photobleaching (FRAP) analysis on AXIN1-mCherry to investigate the structural role of TNKS in the DC. After photobleaching, the fluorescence signals of the AXIN-mCherry puncta in the DMSO control samples recovered quickly ($k = 0.093 \text{ s}^{-1}$, $n = 21$), indicating a dynamic assembly of the DC

Figure 3.

TNKS is required for AXIN puncta formation

(A) AXIN colocalized with TNKS and formed puncta in 293T cells transfected with AXIN1-mCherry plasmid. (B) AXIN distributed diffusely throughout the cytoplasm in TNKS1/2-DKO cells transfected with AXIN1-mCherry plasmid. (C) Introduction of TNKS1 restored AXIN puncta in TNKS1/2-DKO cells transfected with AXIN1-mCherry and TNKS1 plasmids. (D) TNKS2 also induced puncta formation in TNKS1/2-DKO cells transfected with AXIN1-mCherry and TNKS2 plasmids. (E) The catalytic function of TNKS1 is not required for puncta formation as demonstrated in TNKS1/2-DKO cells transfected with AXIN1-mCherry and TNKS1-PD plasmids. (F) Catalytically inactive TNKS2-M1054V also promoted AXIN puncta formation effectively in TNKS1/2-DKO cells transfected with AXIN1-mCherry and TNKS2-M1054V plasmids.

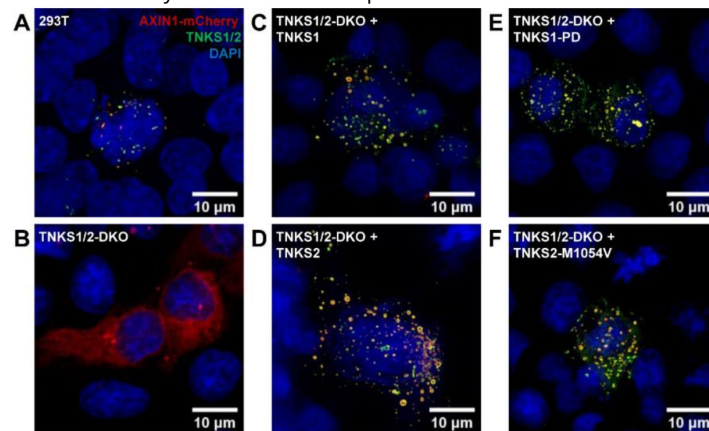
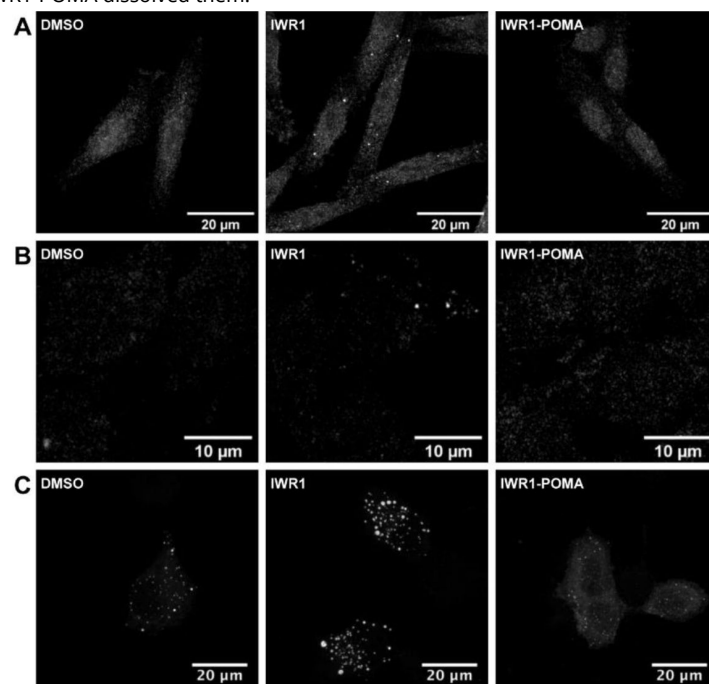


Figure 4.

Chemically induced TNKS accumulation promoted AXIN puncta formation

(A) SW480 cells treated with DMSO, IWR1 (5 μ M) or IWR1-POMA (1 μ M) and stained with anti-AXIN1 antibody. (B) 293T-AXIN1-dsRed-KI cells treated with DMSO, IWR1 (3 μ M) or IWR1-POMA (3 μ M). (C) HeLa cells were transfected with AXIN1-GFP plasmid and then treated with DMSO, IWR1 (3 μ M) or IWR1-POMA (3 μ M). IWR1 promoted the formation of micrometer-sized AXIN puncta whereas IWR1-POMA dissolved them.



(Figure 5B [37](#), [38](#)). However, the fluorescence signals of the AXIN1-mCherry puncta in the IWR1-treated sample recovered slowly ($k = 0.036 \text{ s}^{-1}$, $n = 16$) and plateaued at a much lower level, suggesting that IWR1 rigidified the DC. In contrast, the dynamics of the fluorescence recovery of the few very small AXIN1-mCherry puncta in the IWR1-POMA-treated sample was comparable to that of the DMSO control ($k = 0.090 \text{ s}^{-1}$, $n = 21$). Because both IWR1 and IWR1-POMA stabilized AXIN (Figure 5C [37](#)), the rigidification of the DC by IWR1 is not a result of AXIN accumulation. Instead, as the DC puncta in the IWR1-treated sample was enriched with and that in the IWR1-POMA-treated sample depleted with TNKS, the reduction of the DC dynamics by IWR1 most likely originated from TNKS accumulation. We next performed FRAP analysis on mNeonGreen- β -catenin to examine the functional role of TNKS scaffolding in the DC using the dynamics of β -catenin as a measurement for its turnover rate. In contrast to IWR1-POMA that accelerated the fluorescence recovery of mNeonGreen- β -catenin ($k = 0.046 \text{ s}^{-1}$, $n = 10$), IWR1 reduced the level of recovery by 20% without affecting the kinetics ($k = 0.028 \text{ s}^{-1}$, $n = 16$). These results suggest that TNKS functions in part through cytosolic retention of β -catenin, and TNKS accumulation negatively impacts the catalytic activity of the DC (Figure 5D [37](#)). Collectively, our data support the hypothesis that TNKS scaffolding augments AXIN puncta to limit the ability of the DC to turn over β -catenin.

Tankyrase degradation prevents CRC cell proliferation

Loss of functional APC impairs the DC function and results in constitutive WNT/ β -catenin signaling that drives tumorigenesis and metastasis of CRC. Whereas depletion of both TNKS1 and TNKS2 by shRNA phenocopied APC restoration to prevent tumorigenesis in mice^{[8](#), [39](#)}, TNKSi only showed modest anti-proliferative activities in a limited number of CRC cell lines^{[40](#)}. For example, IWR1 had minimal effects on the proliferation of DLD-1 cells that express truncated forms of APC. Remarkably, IWR1-POMA was able to suppress the formation of DLD-1 cell colonies under both high and low serum conditions (Figure S7A [37](#)). Similarly, the proliferation of SW480 cells that carry a different truncating APC mutation could also be inhibited by IWR1-POMA but not IWR1. The anti-proliferative activity of these molecules also correlated with their ability to suppress WNT/ β -catenin signaling (Figure S7B [37](#)). Mechanistically, several WNT-controlled genes showed differential responses to IWR1 and IWR1-POMA. For example, whereas LEF1 was downregulated in both IWR1 and IWR1-POMA treated samples, cyclin D1 and Aurora kinase A responded to IWR1-POMA only in DLD-1 cells (Figure 1F [37](#) and S7C [37](#) and D [37](#)). This observation is consistent with the ability of IWR1-POMA to better inhibit WNT/ β -catenin signaling. Interestingly, IWR1-POMA also reduced the level of CDK4 while IWR1 had no effect (Figure S7C [37](#) and D [37](#)), which may be explained by CDK4 being a direct target of c-MYC that in turn is a direct target of WNT/ β -catenin. Indeed, IWR1-POMA reduced the level of c-MYC while IWR1 did not (Figure S7D [37](#)). Taken together, targeted degradation of TNKS offers better control of the WNT/ β -catenin pathway activities important to the maintenance of CRC cells than catalytic inhibition.

Several lines of evidence support that the observed growth inhibition by IWR1-POMA is a result of on-target degradation of β -catenin. First, HCT116 cells harboring a mutation in one β -catenin allele that cannot be phosphorylated by the DC showed remarkable resistance to IWR1-POMA (Figure S8A [37](#)). Second, IWR1-POMA had no effect on the level of cytosolic β -catenin in 293T-TNKS1/2-DKO cells that lack both TNKS1 and TNKS2 (Figure S8B [37](#)), indicating that IWR1-POMA downregulated β -catenin in a TNKS-dependent manner. Third, GSPT1/2 degrader CC-90009 (eragidomide)^{[41](#)} had no effect on TNKS and the cytosolic β -catenin levels in DLD-1 cells (Figure S8C [37](#)). Fourth, we have further exposed DLD-1 cells to CC-90009 and established resistant cells (DLD-1R) that lacked the expression of GSPT1/2 (Figure S8D [37](#)). TNKS in DLD-1R remained responsive to IWR1 and IWR1-POMA treatments (Figure S8E [37](#)), and IWR1-POMA was still able to prevent the proliferation of DLD-1R cells effectively (Figure S8F [37](#)). As such, IWR1-POMA suppressed WNT-dependent cancer cell proliferation through degrading TNKS but not GSPT1/2.

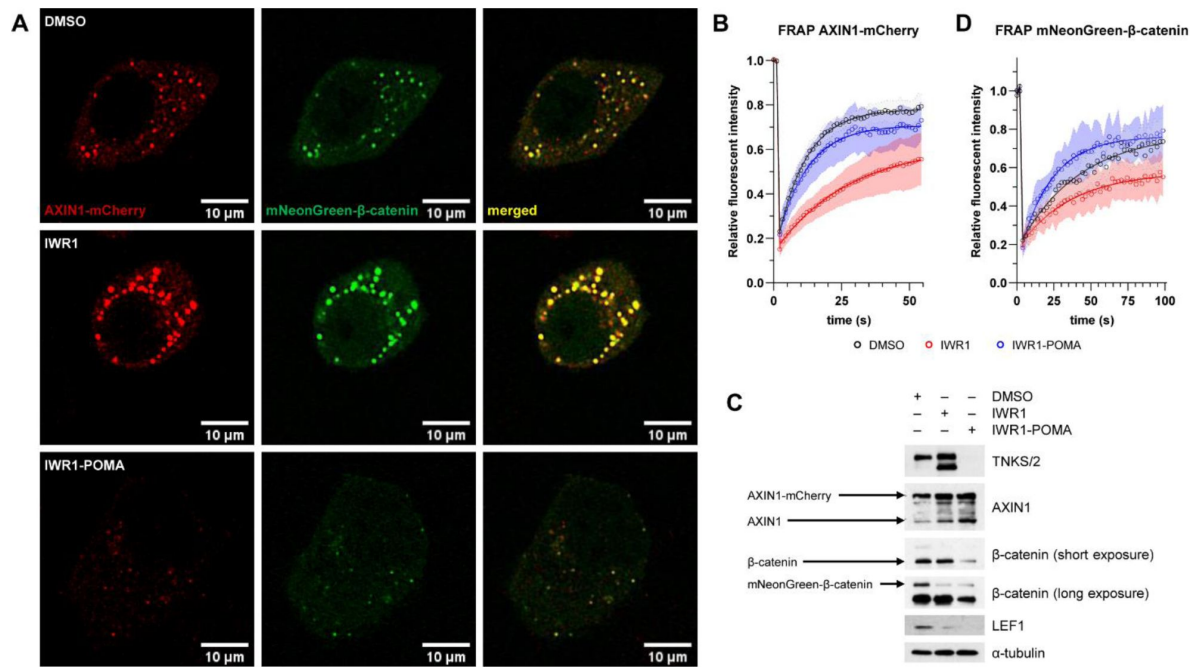


Figure 5.

TNKS accumulation impeded the degradation of β-catenin

(A) HeLa cells were transfected with AXIN1-mCherry and mNeonGreen-β-catenin plasmids and then treated with DMSO, IWR1 (3 μM) or IWR1-POMA (3 μM). IWR1 promoted large AXIN1 puncta formation and IWR1-POMA dissolved the puncta. β-Catenin colocalized with AXIN1, indicating the proper assembly of the DC. (B) FRAP analysis provided support to the hypothesis that TNKS controls the dynamic assembly of the DCs. IWR1 treatment led to a slow recovery of the AXIN1-mCherry fluorescent signal after photobleaching. In contrast, IWR1-POMA treatment did not affect the mobility of AXIN1. The fluorescent recovery curves were fitted to one-phase association model and presented with 95% CI. (C) Western blot analysis of samples corresponding to **Figure 5A** confirmed the accumulation of TNKS by IWR1 and the depletion TNKS by IWR1-POMA. The deeper suppression of the Wnt/β-catenin signaling by IWR1-POMA is also supported by the reduced levels of total β-catenin. (D) FRAP analysis indicated that IWR1 limited the turnover of β-catenin. In contrast, IWR1-POMA accelerated the diffusion rate of β-catenin in the DC.

To investigate the anticancer potential of IWR1-POMA further, we used DLD-1 and SW480 cells to establish 3D spheroids that recapitulated the cell-cell interactions and the hypoxia core of tumors. Consistent with colony formation assays, IWR1 had no effect on the size and the morphology of these spheroids, while IWR1-POMA suppressed their growth effectively (**Figure 6A** and **S9A**). Notably, the outer layer of the spheroids treated by IWR1-POMA had a loose structure incapable of supporting a spherical shape. Instead, they adapted a rather flat architecture with a small core of aggregated cells. Immunostaining showed that the outer layer of these IWR1-POMA-treated spheroids did not contain living cells (**Figure 6B**). The presence of significant DNA breaks in this region (**Figure S9B**) is consistent with the observation that APC restoration induced apoptosis in CRC cells⁴².

We have also evaluated the anticancer effects of IWR1-POMA in a human CRC organoid model that preserved the multicellular identity of the tumor more faithfully than immortalized cancer cells. The PDM-7 patient-derived primary CRC cells harbor truncating mutations in APC, making it susceptible to WNT inhibition. We first confirmed that the PDM-7 organoids maintained the heterogeneous nature of the proliferation and the WNT activity within the organoids (**Figure S9C**). We next confirmed that IWR1-POMA could prevent the formation of PDM-7 organoids whereas IWR1 had little or no effect (**Figure S9D**). We then treated established PDM-7 organoids with IWR1 or IWR1-POMA and found that IWR1 did not affect their growth (**Figure 6C**). In contrast, IWR1-POMA inhibited the growth of these organoids with an apoptotic phenotype (**Figure S9E**). Additionally, IWR1-POMA promoted β -catenin degradation more effectively than IWR1, and IWR1-POMA suppressed the expression of PCNA but IWR1 did not (**Figure 6D**). Collectively, removing TNKS scaffolding is important for achieving anticancer efficacy in CRC by targeting TNKS.

Discussion

In the existing model of WNT/ β -catenin signaling, AXIN is a key component of the DC. It interacts with APC to work as a scaffolding protein for the DC to catalyze β -catenin phosphorylation and ubiquitination. In the cytoplasm, TNKS also associates with the DC and controls the AXIN level through PARdU^{12,14}. As such, AXIN accumulates upon TNKS inhibition, leading to enhanced β -catenin degradation and reduced WNT/ β -catenin pathway activities^{12,15}. Traditionally, the formation of AXIN puncta is viewed as a hallmark of functional DC induction^{16,17}, and the size of the puncta correlates with their catalytic activity—the bigger the puncta, the better their ability to turn over β -catenin. Because TNKS inhibition induces large AXIN puncta, it is generally believed that TNKSi supports the formation of highly functional DCs to accelerate the turnover of β -catenin. However, given the newly discovered role of TNKS scaffolding^{18,19}, the mechanism by which TNKS controls WNT signaling needs refinement.

The DC is a classic example of biomolecular condensates^{9,10}. It containing tens to hundreds of AXIN and APC molecules in a ~1:1 ratio to catalyze β -catenin degradation⁴³. Both AXIN and APC contain extensive intrinsically disordered regions, which are likely important for providing a flexible structure to support catalysis. TNKS interacts with AXIN and APC through multivalent binding in the DC. Based on this work, the catalysis-independent function of TNKS in WNT/ β -catenin signaling originates from its ability to promote maturation^{22,23} of the DC condensates. Specifically, because TNKS aggregates and forms filaments at high concentrations^{31,35,36}, large AXIN puncta containing excess TNKS are rigid and exhibit reduced catalytic activity. In contrast, in the absence of TNKS, the small AXIN complexes are dynamic and catalyze β -catenin degradation more effectively. As such, the size of the AXIN complex does not inform the activity of the DC. This notion is consistent with the previous observation that AXIN2 can also support β -catenin degradation despite having a reduced ability to form puncta⁴⁴. Collectively, our study suggests that TNKS controls WNT/ β -catenin signaling through two independent mechanisms: it regulates AXIN homeostasis through catalytic PARylation¹²; meanwhile it dictates the material

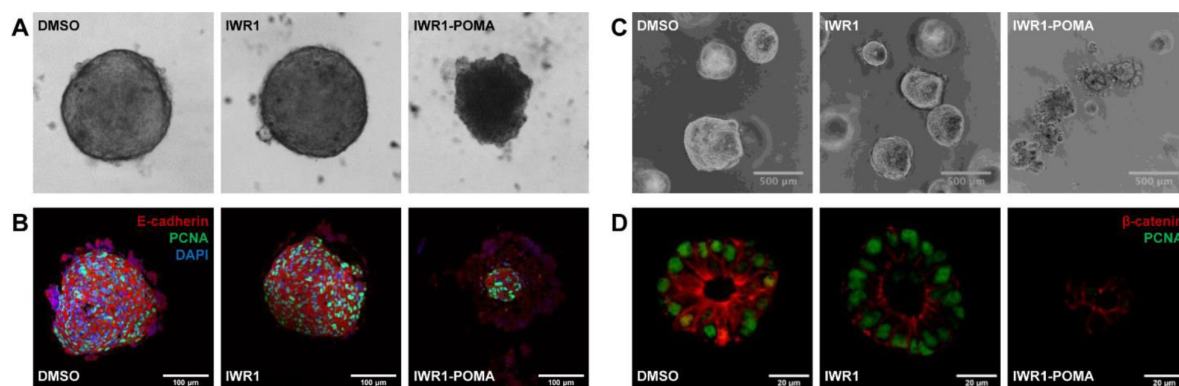


Figure 6.

IWR1-POMA suppressed CRC cell growth

(A) DLD-1 cells were developed into 3D spheroids and then treated with DMSO, IWR1 (5 μ M) or IWR1-POMA (5 μ M) for 10 d. The spheroids treated with IWR1-POMA lost the tight, spherical structure while IWR1 had no effect. (B) Immunostaining showed that the DLD-1 spheroids treated with IWR1-POMA only contained a small core of living cells while those treated with IWR1 remained highly proliferative. (C) IWR1-POMA (1 μ M) suppressed the growth of PDM-7 patient-derived CRC organoids of approximately 200 μ m in diameter with an apoptotic phenotype while IWR1 (1 μ M) had no effect. (D) IWR1-POMA (1 μ M) reduced the level of β -catenin and suppressed the proliferation of PDM-7 organoids grown from single cells significantly more effectively than IWR1 (1 μ M).

properties of the DC through molecular scaffolding. The mechanism by which TNKS antagonizes the DC through molecular scaffolding is also reminiscent of the effect of disheveled (DVL) polymerization on the WNT/ β -catenin signalosome⁴⁵.

The development of TNKSi as an anticancer drug has been discouraged by the lack of significant efficacy in various in vitro and in vivo models. Our data suggest that TNKSd can overcome this limitation by stabilizing AXIN without affecting the DC dynamics. Therefore, TNKSd can provide better suppression of the WNT/ β -catenin-controlled genes important for cancer cell proliferation than TNKSi. In particular, we identify several WNT target genes that are suppressed by TNKSd but not TNKSi. Among them, Aurora A is an oncogene frequently amplified in CRC⁴⁶. Because Aurora A binds to and stabilizes c-MYC that is also dysregulated in many cancers⁴⁷, the ability of IWR1-POMA to downregulate both c-MYC and Aurora A more effectively than IWR1 likely contributed to its improved anti-proliferative activity. Another WNT/ β -catenin downstream target specifically regulated by TNKSd is cyclin D1. The CDK4/cyclin D1 complex is a key regulator of cell cycle, and CDK4 blockade has been explored as a potential strategy to enhance the anticancer efficacy of TNKSi^{48,49}. Interestingly, TNKSd, but not TNKSi, also downregulated CDK4 expression potentially through c-MYC. We speculate that the coordinated action of IWR1-POMA on CDK4 and cyclin D1 additionally contributed to its improved anticancer efficacy. Finally, TNKSi can sensitize cancer cells toward EGFR or MEK inhibition^{50,51}. As a more effective suppressor of Wnt/ β -catenin signaling, IWR1-POMA may provide a better therapeutic window for treating cancers driven by WNT/ β -catenin signaling using a corresponding combinatorial approach.

Materials and methods

Cell lines

HAP1, DLD1, HEK293, 239T, HeLa, HT-29, SW480, and L-Wnt3A cells obtained from Horizon Discovery or ATCC were cultured according to the vendors' recommended procedures. PDM-7 obtained from ATCC were grown into organoid according to the procedures provided by ATCC using the Organoid Growth Kit 1A and cell basement membrane for organoid culture from ATCC.

CRISPR engineering of HAP1 cells

HAP1 cells were transiently transfected with pCRISPaint-NanoLuc-PuroR and pCAS9-mCherry-Frame+2 plasmids and a TNKS1 sgRNA plasmid targeting 5'-TCACTAGGTCTTCTGCTCTGCGG-3' and selected by puromycin treatment. Cells were then cultured in 96 well plates to generate single colonies and sequenced to select for correct incorporation of nanoluciferase at the C-terminus of TNKS1.

DLD-1R cells

DLD-1 cells were cultivated with an increasing concentration of CC-90009 (0.1, 1 and 10 μ M) to confer resistance to GSPT inhibition. Western analysis confirmed the loss of GSPT1 expression in these cells. GSPT2 is a low-abundance protein in DLD-1 and DLD-1R not detectable by Western blot, which is consistent with a reported quantitative proteomic analysis of DLD-1 cells (GSPT1: 102,567 ppb, GSPT2: 2,495 ppb; <https://www.ebi.ac.uk/gxa/experiments/E-PROT-18/Results>)⁵².

Determination of TNKS degradation

HAP1-TNKS1-NanoLuc cells at 30–50% confluence were treated with different concentrations of PROTAC molecules in triplicates for 16 h. The TNKS1 level was quantified by QUANTI-Luc Gold (Invivogen) with normalization to the total protein content. For high-throughput screens, the cells were seeded into 96-well plates at 2×10^4 cells per well with 100 μ L culture media. The PROTAC molecules dissolved in DMSO were placed in T8+ dispensehead cassettes and then introduced by

the TECAN D300e Digital Dispenser with series dilutions under the D300e software control. To determine the DC_{50} value of IWR1-POMA, the dose-response relationship was fitted to a Bayesian Gaussian Processes model in R using code developed by Semenova⁵³.

Immunoblot assays

Cells were washed once with cold PBS and lysed with the SDS lysis buffer (1% SDS, 10 mM HEPES, pH 7.0, 2 mM $MgCl_2$ and 500 U universal nuclease). Subcellular protein fractions were obtained by using the Mem-PER Plus Kit (Thermo Fisher Scientific). Protein concentrations were determined by the BCA assay kit (Thermo Fisher Scientific). A total of 20 μ g of proteins were loaded onto the SDS-PAGE gel and then transferred to a nitrocellulose membrane. Nitrocellulose membranes were then blocked with Tris buffered saline containing 0.1% Tween-20 and 5% milk (Bio-Rad). Membranes were incubated with the primary antibodies overnight at 4 °C and then the secondary antibodies for 1 h at room temperature. The blots were developed with or without enhanced chemiluminescence and were exposed on autoradiograph films or imaged by a BioRad Molecular Imager ChemiDoc XRS System.

Quantitative mass spectrometry

DLD-1 cells at 40% confluence were treated with the drug for 24 h, washed with cold PBS and lysed with the SDS lysis buffer. Protein concentrations were determined by the BCA assay kit (Thermo Fisher Scientific). Two biological replicate samples were prepared with 500 μ g of proteins from each sample, reduced with 2 mM DTT for 10 min and alkylated with 50 mM iodoacetamide for 30 min in dark, and then extracted using methanol-chloroform precipitation. The protein pellets were dissolved in 400 μ L 8 M urea buffer (8 M urea, 50 mM Tris-HCl, 10 mM EDTA, pH 7.5) and digested by Lys-C (Wako, at a 1:100 enzyme/protein ratio) for 2 h. The urea concentration was then reduced to 2 M using freshly made 100 mM ammonium bicarbonate solution. Proteins were subsequently digested with trypsin (Thermo Fisher Scientific, at 1:100 enzyme/protein ratio) overnight. Peptides were desalted with Oasis HLB cartridges (Waters) and re-suspended in 200 mM HEPES (pH 8.5) to a final concentration of 1 μ g/ μ L. For each sample, 100 μ g of peptides were reacted with the corresponding amine-based TMT reagents (Thermo Fisher Scientific) for 1 h. The reactions were quenched with 5% hydroxylamine solution and were combined. Samples were then desalted and a reverse-phase fractionation spin column (Pierce) was used according to the manufacturer's directions to fractionate the sample into 8 fractions. The fractions were dried in a SpeedVac and reconstituted in a 2% acetonitrile, 0.1% TFA buffer followed by injecting onto an Orbitrap Fusion Lumos mass spectrometer coupled to an Ultimate 3000 RSLC-Nano liquid chromatography system. Samples were injected onto a 75 μ m $\times$ 75-cm EasySpray column (Thermo) and eluted with a gradient from 0 $\rightarrow$ 28% buffer B over 180 min. Buffer A contained 2% (v/v) acetonitrile and 0.1% formic acid in water, and buffer B contained 80% (v/v) acetonitrile, 10% (v/v) trifluoroethanol, and 0.1% formic acid in water. The mass spectrometer operated in positive ion mode with a source voltage of 1.8 kV and an ion transfer tube temperature of 275 °C. MS scans were acquired at 120,000 resolution in the Orbitrap and top speed mode was used for SPS-MS3 analysis with a cycle time of 2.5 s. MS2 was performed with CID with a collision energy of 35%. The top 10 fragments were selected for MS3 fragmentation using HCD, with a collision energy of 55%. Dynamic exclusion was set for 25 s after an ion was selected for fragmentation. The raw MS data files were analyzed using Proteome Discoverer v2.4 (Thermo), with peptide identification performed using Sequest HT searching against the human protein database from UniProt. Fragment and precursor tolerances of 10 ppm and 0.6 Da were specified, and three missed cleavages were allowed. Carbamidomethylation of Cys and TMT10plex labelling of N-termini and Lys side-chains were set as fixed modifications, with oxidation of Met set as a variable modification. Protein abundances were determined based on the sum of the signal-to-noise ratios of the reporter ions for all peptides matched to each protein. The false-discovery rate (FDR) cutoff was 1% for all peptides. Statistical analyses (one-way ANOVA and two-sided unpaired t-tests) were performed in R using the peptide data. The MS data have been deposited in the MassIVE repository with the dataset identifier MSV000089098.

Luciferase reporter assays

L-Wnt3A-STF cells, or HeLa, DLD-1, or SW480 cells expressing STF and Renilla-luciferase at 30–50% confluence were treated with the indicated compound for 16 h. The STF-firefly and Renilla luciferase activities were measured by the Dual Luciferase kit (Promega). The WNT/ β -catenin pathway activities were determined by normalizing the STF-firefly activity to the Renilla activity or the total protein level. Statistical analyses (one-way ANOVA and two-sided unpaired t-tests) were performed in GraphPad Prism.

Immunofluorescence microscopy

HeLa cells at 30–50% confluence in a 35 mm dish were transfected with 30 ng of the indicated plasmids for 6 h and then treated with DMSO, IWR (3 μ M) or IWR1-POMA (3 μ M) for 16 h. Images were obtained using a Zeiss LSM 880 Airyscan confocal laser scanning microscope. FRAP analysis was performed at room temperature in the DMEM media without phenol red and with HEPES supplement and the indicated drugs. Defined regions were photobleached at a specific wavelength using the 405 nm or 561 nm laser, and the fluorescence intensities in these regions were collected every 2 s and normalized to the initial intensity before bleaching. The fluorescent signal intensities were determined in ImageJ and analyzed in GraphPad Prism.

2D colony formation assays

DLD-1, SW480, HCT116 or DLD-1R cells were seeded into 6-well plates at 500–2000 cells/well with 2 or 10% FBS in the growth medium. Compounds were added at the indicated concentrations 16 h after seeding, and the growth medium was replenished every 3 d until colony formation was observed. The colonies were fixed with 4% formalin in PBS and stained by a solution of 0.5% crystal violet in 50% methanol solution.

3D spheroid formation assay

DLD-1 or SW480 cells were seeded at 1,000 cells/well into 96-well plates with 10% FBS in the Hyclone X growth medium. For DLD-1 cells, 3D spheroids formed 1 d after seeding. For SW480 cells, rat tail collagen I (Gibco) was added to provide the extracellular matrix, and 3D spheroids formed 5 d after seeding. Compounds were then added at the indicated concentrations. Medium with the indicated drug was replenished every 3 d, and the size of the spheroids were measured at the indicated time by imaging on a Cytation 5 Cell Imaging Multimode Reader and analyzed by Image J. The spheroids at the endpoints were fix with 4% formalin in PBS, sectioned and stained with the indicated markers after parafilm embedding.

CRC organoid formation assay

The organoids were cultured according to the procedures detailed by ATCC. Briefly, dissociated PDM-7 single cells or organoid fragments of about 200 μ M diameter in size were embedded at 5,000,000 cells/mL in the cell base membrane and dispensed as small droplets onto warm 6-well plates. After solidification, the domes were covered with the advanced DMEM:F12 supplemented with HEPES, L-glutamine and B-27, noggin, gastrin, N-acetyl-cysteine, EGF, nicotinamide, A 83-01, and SB 202190 (ATCC formulation 1). The ROCK inhibitor Y-27632 (10 μ M) was also included for the first 3 d of subculture. The organoids were treated with the drug 3 days after seeding for 5 days in the single-cell model and 1 day after seeding for 5 days in the organoid fragment model.

Synthetic procedures

Synthesis of IWR1-TP4-Poma (IWR1-POMA)

To a solution of [1127442-97-0]¹⁵ (351 mg, 0.8 mmol, 1.0 equiv) in methylene chloride (3 mL) was added methanesulfonyl chloride (183 mg, 1.6 mmol, 2.0 equiv), triethylamine (0.33 mL, 2.4 mmol, 3.0 equiv) at 0 °C. After stirring for 2 h, the reaction mixture was concentrated, the residual was re-dissolved in *N,N*-dimethylformamide (3 mL), and sodium azide (208 mg, 3.2 mmol, 4.0 equiv) was then added. After stirring at 50 °C overnight, the reaction was quenched with water and extracted with ethyl acetate for three times. The combined organic layers were washed with brine, dried over sodium sulfate, concentrated, and purified by silica gel flash column chromatography to give azido-IWR2 (335 mg, 90% yield) as a white powder. ¹H NMR (400 MHz, CDCl₃) δ 10.74 (s, 1H), 8.90 (dd, *J* = 6.0, 3.0 Hz, 1H), 8.78 (d, *J* = 4.4 Hz, 1H), 8.09 (d, *J* = 8.1 Hz, 2H), 7.63–7.56 (m, 2H), 7.47 (d, *J* = 4.4 Hz, 1H), 7.36 (d, *J* = 8.2 Hz, 2H), 6.28 (t, *J* = 1.8 Hz, 2H), 4.82 (s, 2H), 3.54–3.50 (m, 2H), 3.46 (dd, *J* = 3.1, 1.6 Hz, 2H), 1.79 (d, *J* = 8.6 Hz, 1H), 1.62 (d, *J* = 8.8 Hz, 1H); MS (ESI) calcd for C₂₆H₂₁N₆O₃ (M+H)⁺ 465.2, found 465.2.

To a solution of azido-IWR2 (4.2 mg, 0.009 mmol, 1.0 equiv) in dimethyl sulfoxide was added [2138439-58-2]⁵⁴ (5 mg, 0.01 mmol, 1.1 equiv), copper(II) sulfate pentahydrate (4.5 mg, 0.018 mmol, 2.0 equiv) and (+)-sodium l-ascorbate (7.2 mg, 0.036 mmol, 4.0 equiv). After stirring at 80 °C overnight, the reaction was quenched with saturated ammonium chloride and extracted with methylene chloride for three times. The combined organic layers were washed with brine, dried over sodium sulfate, concentrated, and purified by silica gel flash column chromatography followed by preparative HPLC to give IWR1-TP4-Poma (IWR1-POMA) (5.4 mg, 63% yield) as a yellow solid. ¹H NMR (400 MHz, CDCl₃) δ 10.75 (s, 1H), 8.94 (dd, *J* = 7.0, 1.9 Hz, 1H), 8.77 (d, *J* = 4.4 Hz, 1H), 8.38 (s, 1H), 8.14–8.07 (m, 2H), 7.73–7.58 (m, 3H), 7.44 (dd, *J* = 8.5, 7.1 Hz, 1H), 7.40–7.34 (m, 2H), 7.07 (dd, *J* = 7.9, 5.7 Hz, 2H), 6.85 (d, *J* = 8.5 Hz, 1H), 6.29 (t, *J* = 1.9 Hz, 2H), 6.02 (s, 2H), 4.87 (dd, *J* = 12.0, 5.4 Hz, 1H), 4.70 (s, 2H), 3.74–3.58 (m, 14H), 3.54 (dq, *J* = 3.5, 1.7 Hz, 2H), 3.49 (dd, *J* = 3.0, 1.6 Hz, 2H), 3.41 (t, *J* = 5.3 Hz, 2H), 2.89–2.66 (m, 3H), 2.14–2.04 (m, 1H), 1.82 (dt, *J* = 8.9, 1.7 Hz, 1H), 1.65 (d, *J* = 8.9 Hz, 1H); MS (ESI) calcd for C₅₀H₅₀N₉O₁₁ (M+H)⁺ 952.4, found 952.3.

Synthesis of IWR1-TP(n)-Poma

Prepared using the same method as described for IWR1-TP4-Poma using pomalidomide derivatives with different PEG chain length.

IWR1-TP1-Poma

7.4 mg, 41% yield. ¹H NMR (400 MHz, CDCl₃) δ 10.73 (s, 1H), 8.95 (dd, *J* = 6.3, 2.7 Hz, 1H), 8.74 (d, *J* = 4.3 Hz, 1H), 8.12 (dt, *J* = 9.1, 1.8 Hz, 3H), 7.71–7.61 (m, 3H), 7.44 (ddd, *J* = 8.4, 7.0, 1.1 Hz, 1H), 7.40–7.36 (m, 2H), 7.07 (d, *J* = 7.1 Hz, 1H), 7.01 (d, *J* = 4.5 Hz, 1H), 6.86 (d, *J* = 8.5 Hz, 1H), 6.29 (q, *J* = 1.6 Hz, 2H), 6.03 (s, 2H), 4.84 (dd, *J* = 12.0, 5.4 Hz, 1H), 4.71 (s, 2H), 3.74 (d, *J* = 5.3 Hz, 2H), 3.54 (dq, *J* = 3.3, 1.6 Hz, 2H), 3.50 (d, *J* = 1.1 Hz, 2H), 3.46 (d, *J* = 6.4 Hz, 2H), 2.89–2.59 (m, 3H), 1.82 (dt, *J* = 8.9, 1.5 Hz, 1H), 1.64 (d, *J* = 8.9 Hz, 1H); MS (ESI) calcd for C₄₄H₃₈N₉O₈ (M+H)⁺ 820.3, found 820.2.

IWR1-TP2-Poma

10.2 mg, 76% yield. ¹H NMR (400 MHz, CDCl₃) δ 10.71 (s, 1H), 8.92 (dd, *J* = 7.4, 1.6 Hz, 1H), 8.72 (d, *J* = 4.4 Hz, 1H), 8.25 (s, 1H), 8.14–8.07 (m, 2H), 7.69–7.57 (m, 3H), 7.43–7.34 (m, 3H), 7.03 (dd, *J* = 5.9, 3.7 Hz, 2H), 6.81 (d, *J* = 8.5 Hz, 1H), 6.29 (t, *J* = 1.9 Hz, 2H), 6.01 (s, 2H), 4.87 (dd, *J* = 12.0, 5.4 Hz, 1H), 4.71 (s, 2H), 3.70 (dd, *J* = 5.9, 3.2 Hz, 2H), 3.66–3.60 (m, 4H), 3.54 (dq, *J* = 3.5, 1.7 Hz, 2H), 3.49 (d, *J* = 1.6 Hz, 4H), 3.35 (t, *J* = 5.3 Hz, 2H), 2.97–2.64 (m, 3H), 1.82 (dt, *J* = 8.9, 1.7 Hz, 1H), 1.64 (d, *J* = 8.7 Hz, 1H); MS (ESI) calcd for C₄₆H₄₂N₉O₉ (M+H)⁺ 864.3, found 864.3.

IWR1-TP3-Poma

11.0 mg, 95 % yield. ^1H NMR (400 MHz, CDCl_3) δ 10.74 (s, 1H), 8.93 (dd, J = 6.1, 2.8 Hz, 1H), 8.78 (d, J = 4.4 Hz, 1H), 8.55 (s, 1H), 8.15–8.06 (m, 2H), 7.70–7.58 (m, 3H), 7.44 (dd, J = 8.5, Hz, 1H), 7.40–7.35 (m, 2H), 7.09 (d, J = 4.4 Hz, 1H), 7.05 (d, J = 7.1 Hz, 1H), 6.84 (d, J = 8.5 Hz, 1H), 6.29 (t, J = 1.9 Hz, 2H), 6.01 (s, 2H), 4.93–4.85 (m, 1H), 4.70 (s, 2H), 3.76–3.59 (m, 10H), 3.54 (dq, J = 3.4, 1.6 Hz, 2H), 3.49 (dd, J = 3.0, 1.6 Hz, 2H), 3.39 (t, J = 5.2 Hz, 2H), 2.95–2.66 (m, 3H), 2.10 (td, J = 9.8, 9.0, 3.3 Hz, 1H), 1.82 (dt, J = 8.9, 1.7 Hz, 1H), 1.65 (d, J = 8.7 Hz, 1H); MS (ESI) calcd for $\text{C}_{48}\text{H}_{46}\text{N}_9\text{O}_{10}$ ($\text{M}+\text{H}$) $^+$ 908.3, found 908.3.

IWR1-TP5-Poma

7.6 mg, 78% yield. ^1H NMR (400 MHz, CDCl_3) δ 10.76 (s, 1H), 8.95 (dd, J = 7.5, 1.4 Hz, 1H), 8.77 (d, J = 4.4 Hz, 1H), 8.51 (s, 1H), 8.15–8.06 (m, 2H), 7.77–7.62 (m, 3H), 7.46 (dd, J = 8.5, Hz, 1H), 7.42–7.33 (m, 2H), 7.21–7.14 (m, 2H), 7.10–7.03 (m, 2H), 6.87 (d, J = 8.5 Hz, 1H), 6.29 (t, J = 1.8 Hz, 2H), 6.03 (s, 2H), 4.88 (dd, J = 11.9, 5.4 Hz, 1H), 4.68 (s, 2H), 3.75–3.58 (m, 18H), 3.54 (dq, J = 3.4, 1.7 Hz, 2H), 3.48 (dd, J = 3.0, 1.6 Hz, 2H), 3.41 (t, J = 5.2 Hz, 2H), 2.91–2.63 (m, 3H), 2.10 (dt, J = 10.5, 4.1 Hz, 1H), 1.82 (dt, J = 8.9, 1.7 Hz, 1H), 1.65 (dd, J = 8.8, 1.6 Hz, 1H); MS (ESI) calcd for $\text{C}_{52}\text{H}_{54}\text{N}_9\text{O}_{12}$ ($\text{M}+\text{H}$) $^+$ 996.4, found 996.4.

IWR1-TP6-Poma

13.1 mg, 91% yield. ^1H NMR (400 MHz, CDCl_3) δ 10.75 (s, 1H), 8.94 (d, J = 7.5 Hz, 1H), 8.77 (d, J = 4.3 Hz, 1H), 8.61 (s, 1H), 8.11 (d, J = 8.4 Hz, 2H), 7.84–7.60 (m, 3H), 7.44 (dd, J = 8.5, Hz, 1H), 7.37 (d, J = 8.3 Hz, 2H), 7.06 (dd, J = 9.4, 5.7 Hz, 2H), 6.86 (d, J = 8.5 Hz, 1H), 6.29 (t, J = 1.9 Hz, 2H), 6.04 (s, 2H), 4.88 (dd, J = 11.7, 5.4 Hz, 1H), 4.69 (s, 2H), 3.68 (t, J = 5.1 Hz, 4H), 3.64–3.56 (m, 20H), 3.55–3.51 (m, 2H), 3.48 (d, J = 2.1 Hz, 2H), 3.41 (t, J = 5.3 Hz, 2H), 2.89–2.69 (m, 3H), 2.14–2.04 (m, 1H), 1.85–1.77 (m, 1H), 1.64 (d, J = 8.9 Hz, 1H); MS (ESI) calcd for $\text{C}_{54}\text{H}_{58}\text{N}_9\text{O}_{13}$ ($\text{M}+\text{H}$) $^+$ 1040.4, found 1040.4.

Synthesis of IWR1-P(n)-Poma

To a solution of [2472645-01-3]⁵⁵ (1.0 equiv) in methylene chloride was added amino-PEG(n)-pomalidomide (1.05 equiv) followed by 4Å molecular sieves. After stirring at 23 °C overnight, sodium triacetoxyborohydride (20 equiv) was added and the reaction was stirred for 4 h before quenched with water. The mixture was extracted with methylene chloride for three times. The combined organic layers were washed with brine, dried over sodium sulfate, concentrated, and purified by silica gel flash column chromatography followed by preparative HPLC to give IWR1-P(n)-Poma as a yellow solid.

IWR1-P1-Poma

16.0 mg, 78% yield. ^1H NMR (500 MHz, CDCl_3) δ 10.48 (s, 1H), 9.09 (s, 1H), 8.67 (t, J = 6.1 Hz, 2H), 7.97 (d, J = 8.1 Hz, 2H), 7.55 (d, J = 8.5 Hz, 2H), 7.43 (t, J = 8.0 Hz, 1H), 7.36 (t, J = 7.7 Hz, 1H), 7.32 (d, J = 8.1 Hz, 2H), 6.96 (d, J = 7.1 Hz, 1H), 6.76 (d, J = 8.6 Hz, 1H), 6.27 (t, J = 1.9 Hz, 2H), 4.74 (dd, J = 13.1, 5.3 Hz, 1H), 4.59 (s, 2H), 3.74 (s, 2H), 3.62 (t, J = 5.0 Hz, 2H), 3.54–3.49 (m, 2H), 3.47 (s, 4H), 3.34 (d, J = 5.6 Hz, 2H), 3.28–3.19 (m, 2H), 2.79–2.38 (m, 3H), 1.93 (d, J = 12.4 Hz, 1H), 1.80 (dt, J = 8.9, 1.7 Hz, 1H), 1.63 (d, J = 8.8 Hz, 1H); MS (ESI) calcd for $\text{C}_{43}\text{H}_{40}\text{N}_7\text{O}_8$ ($\text{M}+\text{H}$) $^+$ 782.3, found 782.2.

IWR1-P2-Poma

13.1 mg, 56% yield. ^1H NMR (500 MHz, CDCl_3) δ 10.53 (s, 1H), 9.39–9.14 (m, 1H), 8.71 (dd, J = 23.5, 5.9 Hz, 2H), 8.12–7.91 (m, 2H), 7.60 (t, J = 7.4 Hz, 2H), 7.49 (t, J = 8.2 Hz, 1H), 7.36 (d, J = 8.0 Hz, 2H), 7.32–7.23 (m, 2H), 7.16 (dd, J = 12.7, 7.1 Hz, 1H), 6.95 (d, J = 7.0 Hz, 1H), 6.60 (d, J = 8.5 Hz, 1H), 6.29 (t, J = 1.9 Hz, 2H), 4.91 (dd, J = 12.7, 5.4 Hz, 1H), 4.69 (s, 2H), 3.85 (t, J = 4.8 Hz, 2H), 3.68 (d, J = 4.3 Hz, 3H),

3.63 (dt, $J = 10.2, 4.1$ Hz, 4H), 3.58–3.45 (m, 6H), 3.39 (s, 2H), 3.22 (t, $J = 5.1$ Hz, 2H), 2.85–2.67 (m, 2H), 2.62 (td, $J = 12.8, 4.7$ Hz, 1H), 2.08–1.96 (m, 1H), 1.82 (dt, $J = 9.0, 1.8$ Hz, 1H), 1.65 (d, $J = 8.8$ Hz, 1H); MS (ESI) calcd for $C_{45}H_{44}N_7O_9$ ($M+H$)⁺ 826.3, found 826.3.

IWR1-P3-Poma

13.2 mg, 72% yield. 1H NMR (500 MHz, $CDCl_3$) δ 10.65 (s, 1H), 9.23 (s, 1H), 8.83 (d, $J = 7.6$ Hz, 1H), 8.75 (d, $J = 4.0$ Hz, 1H), 8.10–8.01 (m, 2H), 7.66 (t, $J = 7.3$ Hz, 2H), 7.56 (t, $J = 8.2$ Hz, 1H), 7.45–7.31 (m, 3H), 7.00 (d, $J = 7.1$ Hz, 1H), 6.74 (d, $J = 8.6$ Hz, 1H), 6.29 (d, $J = 1.8$ Hz, 2H), 4.92–4.82 (m, 1H), 4.69 (s, 2H), 3.79 (s, 2H), 3.67–3.51 (m, 14H), 3.51–3.43 (m, 4H), 3.28 (s, 2H), 2.89–2.59 (m, 3H), 2.05 (q, $J = 9.1, 7.1$ Hz, 1H), 1.85–1.77 (m, 1H), 1.64 (d, $J = 8.9$ Hz, 1H); MS (ESI) calcd for $C_{47}H_{48}N_7O_{10}$ ($M+H$)⁺ 870.3, found 870.3.

IWR1-P4-Poma

15.6 mg, 68% yield. 1H NMR (500 MHz, $CDCl_3$) δ 10.66 (s, 1H), 9.02 (s, 1H), 8.85 (d, $J = 7.7$ Hz, 1H), 8.76 (d, $J = 3.9$ Hz, 1H), 8.07 (d, $J = 8.2$ Hz, 2H), 7.77–7.65 (m, 2H), 7.56 (t, $J = 8.1$ Hz, 1H), 7.36 (d, $J = 8.0$ Hz, 2H), 7.32 (t, $J = 7.8$ Hz, 1H), 6.97 (d, $J = 7.0$ Hz, 1H), 6.64 (d, $J = 8.5$ Hz, 1H), 6.29 (d, $J = 2.0$ Hz, 2H), 4.87 (q, $J = 5.7$ Hz, 1H), 4.71 (s, 2H), 3.80 (s, 2H), 3.68–3.40 (m, 22H), 3.32 (s, 2H), 3.11 (d, $J = 5.3$ Hz, 2H), 2.88–2.58 (m, 3H), 2.12–2.00 (m, 1H), 1.81 (d, $J = 8.9$ Hz, 1H), 1.64 (d, $J = 8.8$ Hz, 1H); MS (ESI) calcd for $C_{49}H_{52}N_7O_{11}$ ($M+H$)⁺ 914.4, found 914.3.

IWR1-P5-Poma

13.6 mg, 53% yield. 1H NMR (500 MHz, $CDCl_3$) δ 10.72 (s, 1H), 8.89 (d, $J = 7.7$ Hz, 1H), 8.86 (s, 2H), 8.81 (d, $J = 4.1$ Hz, 1H), 8.10 (d, $J = 8.0$ Hz, 2H), 7.74 (dd, $J = 13.1, 6.4$ Hz, 2H), 7.60 (t, $J = 8.1$ Hz, 1H), 7.37 (dd, $J = 8.4, 3.9$ Hz, 3H), 7.24 (d, $J = 7.6$ Hz, 1H), 7.16 (dd, $J = 12.6, 7.1$ Hz, 1H), 7.01 (d, $J = 7.1$ Hz, 1H), 6.74 (d, $J = 8.5$ Hz, 1H), 6.29 (d, $J = 1.9$ Hz, 2H), 4.99–4.83 (m, 1H), 4.75 (s, 2H), 3.80 (s, 2H), 3.70–3.41 (m, 24H), 3.32 (s, 2H), 3.26 (t, $J = 4.2$ Hz, 2H), 2.76 (ddd, $J = 61.7, 10.7, 4.5$ Hz, 3H), 2.08 (dq, $J = 7.6, 4.1, 2.7$ Hz, 1H), 1.81 (d, $J = 9.0$ Hz, 1H), 1.64 (d, $J = 8.8$ Hz, 1H); MS (ESI) calcd for $C_{51}H_{56}N_7O_{12}$ ($M+H$)⁺ 958.4, found 958.4.

IWR1-P6-Poma

9.2 mg, 40% yield. 1H NMR (500 MHz, $CDCl_3$) δ 10.74 (s, 1H), 8.90 (d, $J = 7.5$ Hz, 2H), 8.83 (d, $J = 4.1$ Hz, 1H), 8.11 (d, $J = 7.9$ Hz, 2H), 7.76 (dd, $J = 11.8, 6.5$ Hz, 2H), 7.60 (t, $J = 8.1$ Hz, 1H), 7.38 (dd, $J = 8.6, 2.2$ Hz, 3H), 7.02 (d, $J = 7.1$ Hz, 1H), 6.73 (d, $J = 8.5$ Hz, 1H), 6.29 (t, $J = 1.8$ Hz, 2H), 4.89 (q, $J = 5.5, 5.1$ Hz, 1H), 4.77 (t, $J = 9.9$ Hz, 2H), 3.83 (s, 2H), 3.68–3.44 (m, 28H), 3.35 (s, 2H), 3.28 (d, $J = 5.1$ Hz, 2H), 2.93–2.62 (m, 3H), 2.18–2.01 (m, 1H), 1.85–1.79 (m, 1H), 1.65 (d, $J = 8.9$ Hz, 1H); MS (ESI) calcd for $C_{53}H_{60}N_7O_{13}$ ($M+H$)⁺ 1002.4, found 1002.4.

Synthesis of IWR1-P(n)-VHL

Prepared as a white powder using the same method as described for IWR1-P(n)-Poma using amino-PEG(n)-VH032 derivatives⁵⁶ with different PEG chain length.

IWR1-P1-VHL

2.4 mg, 9% yield. 1H NMR (400 MHz, $CDCl_3$) δ 10.75 (s, 1H), 8.97 (d, $J = 7.7$ Hz, 1H), 8.83 (d, $J = 4.4$ Hz, 1H), 8.75 (s, 1H), 8.20–8.00 (m, 2H), 7.89 (d, $J = 4.5$ Hz, 1H), 7.84 (d, $J = 8.6$ Hz, 1H), 7.71 (t, $J = 8.1$ Hz, 1H), 7.41–7.32 (m, 2H), 7.29 (s, 4H), 6.66 (d, $J = 8.7$ Hz, 1H), 6.29 (t, $J = 1.9$ Hz, 2H), 4.92 (s, 2H), 4.68–4.43 (m, 4H), 4.28 (dd, $J = 15.2, 5.2$ Hz, 1H), 4.02 (d, $J = 11.4$ Hz, 1H), 3.87 (s, 1H), 3.76 (s, 3H), 3.59 (d, $J = 11.0$ Hz, 1H), 3.55 (s, 3H), 3.49 (t, $J = 2.3$ Hz, 2H), 3.30 (q, $J = 18.2, 15.0$ Hz, 5H), 2.49 (s, 3H), 1.82 (d, $J = 8.9$ Hz, 1H), 1.65 (d, $J = 9.0$ Hz, 2H), 0.96 (s, 9H); MS (ESI) calcd for $C_{53}H_{59}N_8O_8S$ ($M+H$)⁺ 967.4, found 967.4.

IWR1-P2-VHL

5.3 mg, 18% yield. ^1H NMR (400 MHz, CDCl_3) δ 10.76 (t, J = 2.8 Hz, 1H), 8.96 (q, J = 4.6, 3.7 Hz, 2H), 8.85 (q, J = 3.2, 2.3 Hz, 1H), 8.10 (dt, J = 8.7, 2.7 Hz, 2H), 7.99–7.58 (m, 5H), 7.42–7.33 (m, 4H), 7.31 (t, J = 2.8 Hz, 2H), 7.02 (d, J = 8.6 Hz, 1H), 6.47–6.13 (m, 2H), 4.90 (d, J = 5.4 Hz, 2H), 4.58 (td, J = 19.0, 16.8, 11.0 Hz, 5H), 4.31 (dd, J = 12.0, 8.0 Hz, 1H), 3.99 (d, J = 11.2 Hz, 1H), 3.85 (s, 3H), 3.70–3.60 (m, 7H), 3.55 (s, 3H), 3.50 (t, J = 3.2 Hz, 4H), 3.37–3.23 (m, 3H), 2.61–2.36 (m, 6H), 2.25 (d, J = 26.9 Hz, 3H), 1.82 (d, J = 9.0 Hz, 1H), 1.70–1.53 (m, 1H), 1.32 (td, J = 7.3, 3.4 Hz, 4H), 0.97 (d, J = 3.0 Hz, 10H); MS (ESI) calcd for $\text{C}_{55}\text{H}_{63}\text{N}_8\text{O}_9\text{S}$ ($\text{M}+\text{H}$) $^+$ 1011.4, found 1011.4.

IWR1-P3-VHL

4.7 mg, 23% yield. ^1H NMR (400 MHz, CDCl_3) δ 10.76 (s, 1H), 9.15 (s, 1H), 8.96 (d, J = 7.5 Hz, 1H), 8.89 (d, J = 4.1 Hz, 1H), 8.10 (dd, J = 8.6, 2.1 Hz, 2H), 7.95–7.87 (m, 1H), 7.79 (d, J = 8.6 Hz, 1H), 7.70 (t, J = 8.2 Hz, 1H), 7.38 (d, J = 7.9 Hz, 4H), 7.32 (d, J = 7.7 Hz, 2H), 6.29 (s, 2H), 4.91 (s, 2H), 4.76–4.39 (m, 4H), 3.86 (s, 2H), 3.80–3.41 (m, 17H), 3.37–3.13 (m, 2H), 2.54 (d, J = 2.0 Hz, 3H), 2.43 (s, 2H), 2.27 (s, 2H), 1.82 (d, J = 8.9 Hz, 1H), 1.65 (d, J = 8.8 Hz, 1H), 1.37–1.29 (m, 3H), 0.96 (s, 9H); MS (ESI) calcd for $\text{C}_{57}\text{H}_{67}\text{N}_8\text{O}_{10}\text{S}$ ($\text{M}+\text{H}$) $^+$ 1055.5, found 1055.4.

IWR1-P4-VHL

3.3 mg, 32% yield. ^1H NMR (400 MHz, CDCl_3) δ 10.72 (s, 1H), 8.89 (d, J = 7.7 Hz, 1H), 8.85–8.77 (m, 2H), 8.02 (d, J = 8.2 Hz, 2H), 7.81 (dd, J = 20.6, 6.6 Hz, 2H), 7.61 (t, J = 8.1 Hz, 1H), 7.36–7.23 (m, 6H), 6.91 (d, J = 10.9 Hz, 1H), 6.21 (t, J = 1.9 Hz, 2H), 4.81 (s, 2H), 4.66–4.18 (m, 5H), 3.82 (d, J = 20.1 Hz, 3H), 3.65–3.37 (m, 23H), 3.25 (s, 2H), 2.46 (s, 3H), 1.75 (d, J = 8.7 Hz, 1H), 1.57 (d, J = 8.9 Hz, 1H), 1.29 (t, J = 7.0 Hz, 3H), 0.86 (s, 9H); MS (ESI) calcd for $\text{C}_{59}\text{H}_{71}\text{N}_8\text{O}_{11}\text{S}$ ($\text{M}+\text{H}$) $^+$ 1099.5, found 1099.5.

IWR1-P5-VHL

6.9 mg, 32% yield. ^1H NMR (400 MHz, CDCl_3) δ 10.70 (s, 1H), 8.94 (s, 1H), 8.90 (d, J = 7.7 Hz, 1H), 8.82 (d, J = 4.0 Hz, 1H), 8.04 (d, J = 8.1 Hz, 2H), 7.88–7.85 (m, 1H), 7.76 (d, J = 8.6 Hz, 1H), 7.63 (t, J = 8.1 Hz, 1H), 7.30 (dd, J = 12.7, 7.1 Hz, 7H), 6.22 (d, J = 2.2 Hz, 2H), 4.85 (s, 2H), 4.72–4.35 (m, 4H), 4.27 (d, J = 15.0 Hz, 1H), 3.82 (s, 2H), 3.72–3.28 (m, 21H), 3.24 (d, J = 7.6 Hz, 3H), 2.46 (s, 3H), 1.75 (d, J = 8.9 Hz, 1H), 1.58 (d, J = 8.9 Hz, 1H), 1.28 (t, J = 6.8 Hz, 3H), 0.88 (s, 9H); MS (ESI) calcd for $\text{C}_{61}\text{H}_{75}\text{N}_8\text{O}_{12}\text{S}$ ($\text{M}+\text{H}$) $^+$ 1143.5, found 1143.4.

IWR1-P6-VHL

6.7 mg, 20% yield. ^1H NMR (400 MHz, CDCl_3) δ 10.71 (s, 1H), 9.02 (s, 1H), 8.90 (d, J = 7.7 Hz, 1H), 8.83 (d, J = 4.3 Hz, 1H), 8.10–7.99 (m, 2H), 7.86 (d, J = 4.4 Hz, 1H), 7.76 (d, J = 8.5 Hz, 1H), 7.64 (t, J = 8.1 Hz, 1H), 7.38–7.25 (m, 7H), 6.22 (t, J = 1.9 Hz, 2H), 4.87 (s, 2H), 4.50 (ddd, J = 35.0, 24.8, 16.4 Hz, 4H), 4.27 (dd, J = 15.2, 4.9 Hz, 1H), 4.02 (d, J = 11.3 Hz, 1H), 3.82 (d, J = 5.4 Hz, 2H), 3.68–3.38 (m, 32H), 3.27 (d, J = 7.3 Hz, 2H), 2.47 (s, 3H), 1.75 (dt, J = 9.0, 1.7 Hz, 1H), 1.58 (d, J = 8.9 Hz, 1H), 1.30 (t, J = 7.0 Hz, 3H), 0.89 (s, 9H); MS (ESI) calcd for $\text{C}_{63}\text{H}_{79}\text{N}_8\text{O}_{13}\text{S}$ ($\text{M}+\text{H}$) $^+$ 1187.5, found 1187.5.

Synthesis of IWR6-P(n)-Poma and IWR6-P(n)-VHL

Prepared using the same methods as described for IWR1-P(n)-Poma and IWR1-P(n)-VHL using [1801530-64-2]⁵⁷.

Synthesis of IWR1-C6-Poma

Prepared using the same methods as described for IWR1-P(n)-Poma using [2093386-50-4]⁵⁸. MS (ESI) calcd for $\text{C}_{45}\text{H}_{44}\text{N}_7\text{O}_7$ ($\text{M}+\text{H}$) $^+$ 794.3, found 794.3.

Synthesis of IWR1-R-OLena

To a solution of [2472645-01-3]⁵⁵ (40 mg, 0.09 mmol, 1.0 equiv) in methylene chloride (0.5 mL) was added [852180-47-3]⁵⁹ (28 mg, 0.095 mmol, 1.05 equiv) followed by 4Å molecular sieves (400 mg). After stirring at 23 °C overnight, sodium triacetoxyborohydride (48 mg, 0.23 mmol, 2.5 equiv) was added and the reaction was stirred for 4 h before quenched with water. The mixture was extracted with methylene chloride for three times. The combined organic layers were washed with brine, dried over sodium sulfate, concentrated, and purified by silica gel flash column chromatography followed by preparative HPLC to give IWR1-RL (45 mg, 69% yield) as a white powder. ¹H NMR (400 MHz, CDCl₃) δ 10.82 (s, 1H), 8.89 (d, *J* = 7.8 Hz, 1H), 8.78 (dd, *J* = 4.4, 1.2 Hz, 1H), 8.16–8.03 (m, 2H), 7.67–7.62 (m, 1H), 7.61–7.53 (m, 2H), 7.41–7.34 (m, 2H), 7.27 (d, *J* = 8.0 Hz, 2H), 6.95–6.82 (m, 2H), 6.29 (q, *J* = 1.7 Hz, 2H), 4.86 (s, 4H), 4.29 (s, 2H), 3.88 (s, 2H), 3.68–3.36 (m, 8H), 3.13 (t, *J* = 5.2 Hz, 4H), 2.04 (d, *J* = 1.2 Hz, 4H), 1.81 (dd, *J* = 8.9, 1.7 Hz, 1H), 1.64 (d, *J* = 8.8 Hz, 1H), 1.48 (d, *J* = 1.3 Hz, 9H), 1.25 (td, *J* = 7.1, 1.2 Hz, 1H); MS (ESI) calcd for C₄₂H₄₅N₆O₅ (M+H)⁺ 713.3, found 713.4.

To a solution of IWR1-RL (17 mg, 0.024 mmol, 1.0 equiv) in methylene chloride (0.6 mL) was added trifluoroacetic acid (0.3 L). After stirring at 40 °C overnight, the reaction mixture was concentrated and redissolved in acetonitrile (0.5 mL) before [1323407-86-8]⁶⁰ (10.8 mg, 0.024 mmol, 1.0 equiv) and *N,N*-diisopropylethylamine (13 µL, 0.073 mmol, 3.0 equiv) was added. After stirring at 23 °C overnight, water was added the reaction mixture was extracted with ethyl acetate for three times. The combined organic layers were washed with brine, dried over sodium sulfate, concentrated, and purified by silica gel flash column chromatography to give IWR1-R-OLena (5.5 mg, 23% yield) as a white powder. ¹H NMR (400 MHz, CD₃OD) δ 9.00 (d, *J* = 4.5 Hz, 1H), 8.90 (dd, *J* = 7.3, 1.6 Hz, 1H), 8.22–8.03 (m, 2H), 7.83–7.56 (m, 8H), 7.53–7.39 (m, 7H), 7.31 (dd, *J* = 8.1, 0.9 Hz, 1H), 7.16–7.09 (m, 2H), 6.32 (t, *J* = 1.9 Hz, 2H), 5.35 (s, 2H), 5.18 (dd, *J* = 13.3, 5.2 Hz, 1H), 4.64–4.36 (m, 6H), 3.60 (dd, *J* = 3.0, 1.6 Hz, 2H), 3.48 (dq, *J* = 3.4, 1.7 Hz, 2H), 2.94 (ddd, *J* = 18.5, 13.5, 5.5 Hz, 1H), 2.81 (ddd, *J* = 17.6, 4.7, 2.4 Hz, 1H), 2.52 (qd, *J* = 13.2, 4.6 Hz, 1H), 2.21 (dtd, *J* = 12.7, 5.3, 2.5 Hz, 1H), 1.81 (dt, *J* = 8.8, 1.7 Hz, 1H), 1.73 (d, *J* = 8.8 Hz, 1H), 1.32 (s, 1H); MS (ESI) calcd for C₅₈H₅₅N₈O₇ (M+H)⁺ 975.4, found 975.4.

Supplementary figures

Figure S1.

Crystal structures used to guide the design of PROTAC molecules

(A) The crystal structure of TNKS1 with IWR1-exo (PDB 4OA7). (B) The crystal structure of TNKS1 with XAV939 (PDB 3UH4). (C) The crystal structure of TNKS2 with IWR1 (PDB 3UA9). (D) The crystal structure of TNKS2 with XAV939 (PDB 3KR8).

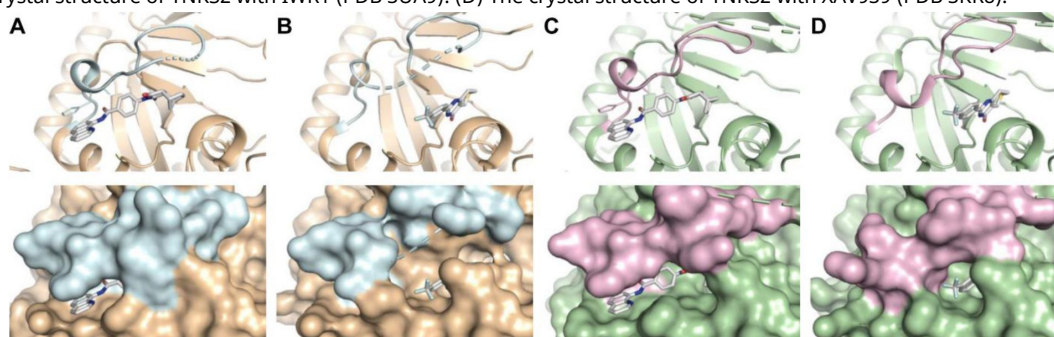


Figure S2.

Identification of active PROTAC molecules using CRISPR engineered HAP1 cells expressing a TNKS1-NanoLuc fusion protein

(A) The chemical structures of the PROTAC molecules. (B) The schematic diagrams of the domain structures of TNKS1, TNKS2 and TNKS1-NanoLuc. (C) The relative abundance of the endogenous TNKS1 was measured by the luciferase activity upon treating HAP1-TNKS1-NanoLuc cells with IWR1-P(n)-VHL, IWR6-P(n)-VHL, IWR1-P(n)-Poma, IWR6-P(n)-Poma, or IWR1-TP(n)-Poma. IWR1-R-Olena bearing a rigid linker of length and polarity comparable to IWR1-P4-Poma and IWR1-P5-Poma alleviated the hook effect but was less effectively in promoting TNKS1 degradation. Removing the oxygen atom from the linker of IWR1-P1-Poma gave IWR1-C6-Poma with a more hydrophobic linker, but there was no improvement in the degradation efficacy.

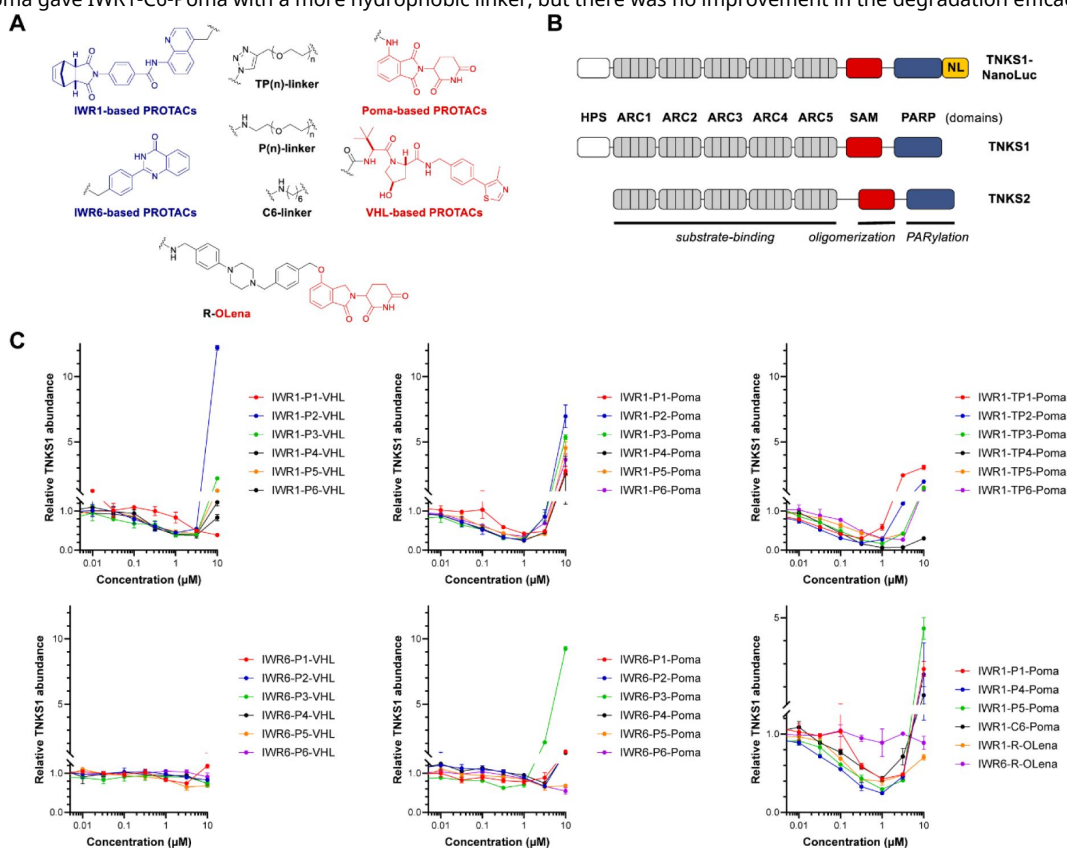


Figure S3.

Additional characterization of IWR1-POMA

(A) TNKS did not recover in DLD-1 cells at least 36 h after removing IWR1-POMA. (B) IWR1 (3 μ M) and pomalidomide (3 μ M) prevented the degradation of TNKS by IWR1-POMA (3 μ M) in DLD-1 cells. (C) IWR1, pomalidomide, and MG132 blocked the degradation of TNKS by IWR1-POMA in HAP1-TNKS1-NanoLuc cells. (D) IWR1-POMA promoted TNKS degradation while IWR1 induced TNKS accumulation in SW480, HT-29 and HeLa cells.

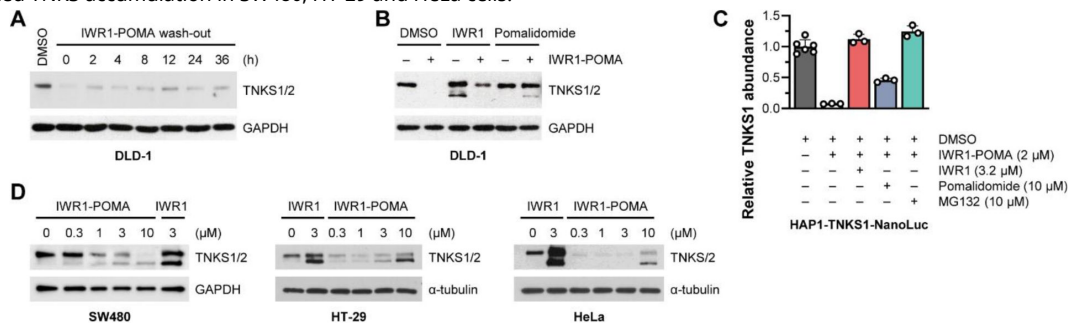
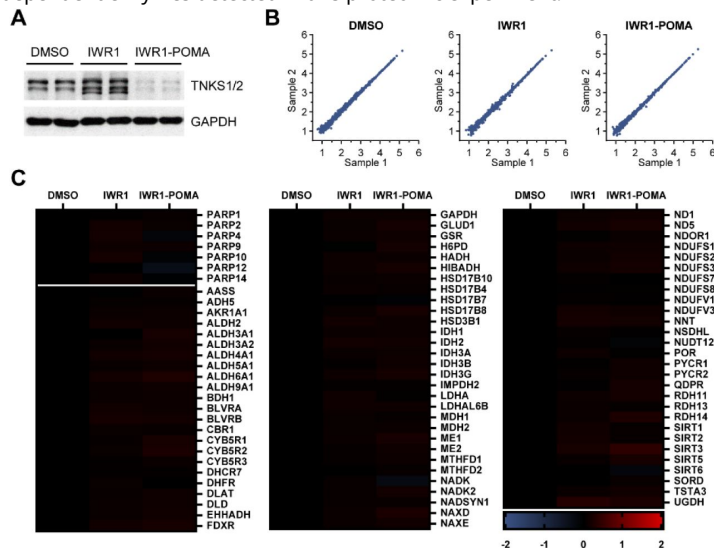


Figure S4.

Proteomic analysis of DLD-1 cells treated with DMSO, IWR1 or IWR1-POMA

(A) Western blot analysis of samples corresponding to Figure 1 E and 1F confirmed the accumulation of TNKS1/2 by IWR1 and the depletion by IWR1-POMA. (B) Correlation analysis showed high reproducibility between the two biological repeats. (C) IWR1-POMA selectively degraded TNKS without inducing appreciable perturbations to 7 other PARP family member proteins and 79 NAD(P)-dependent enzymes detected in this proteomic experiment.



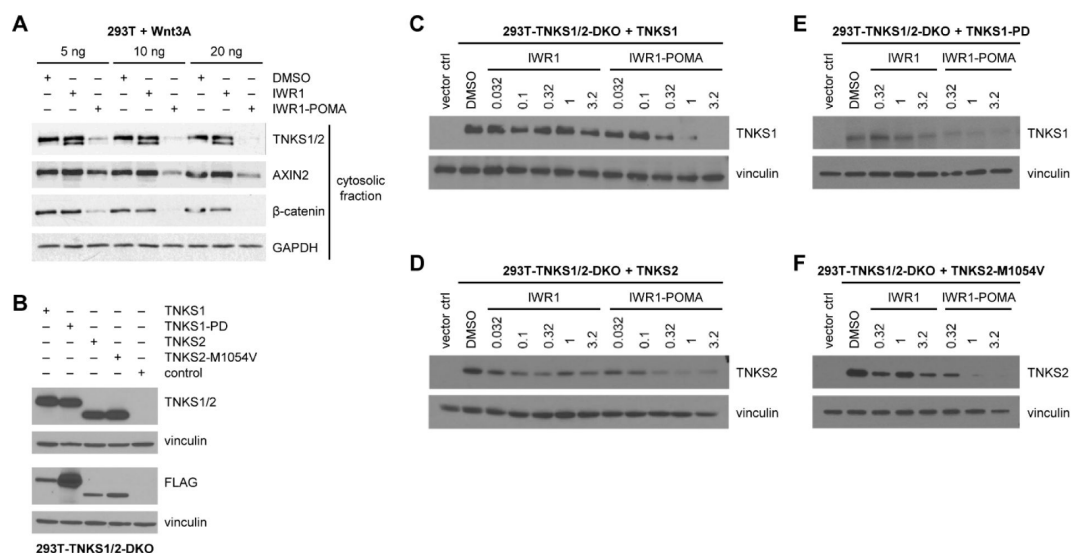


Figure S5.

Degradation of TNKS by IWR1-POMA in 293T cells

(A) 293T cells were transfected with different doses of Wnt3A plasmid and then treated with DMSO, IWR1 (3 μ M) or IWR1-POMA (3 μ M). The cytosolic fraction of the cell lysates was then examined by Western blot. IWR1-POMA promoted a more complete degradation of β -catenin than IWR1 and reduced the level of the WNT target AXIN2. (B) Western blot analysis confirmed the lack of TNKS expression in TNKS1/2-DKO cells and validated the expression of FLAG-TNKS1, 3 $\times$ FLAG-TNKS1-PD, FLAG-TNKS2 and FLAG-TNKS2-M1054V after transfection. (C) Western blot analysis of samples corresponding to [Figure 2A](#) confirmed the degradation of TNKS1 by IWR1-POMA. (D) Western blot analysis of samples corresponding to [Figure 2B](#) confirmed the degradation of TNKS2 by IWR1-POMA. (E) Western blot analysis of samples corresponding to [Figure 2C](#) confirmed the degradation of TNKS1-PD by IWR1-POMA. (F) Western blot analysis of samples corresponding to [Figure 2D](#) confirmed the degradation of TNKS2-M1054V by IWR1-POMA.

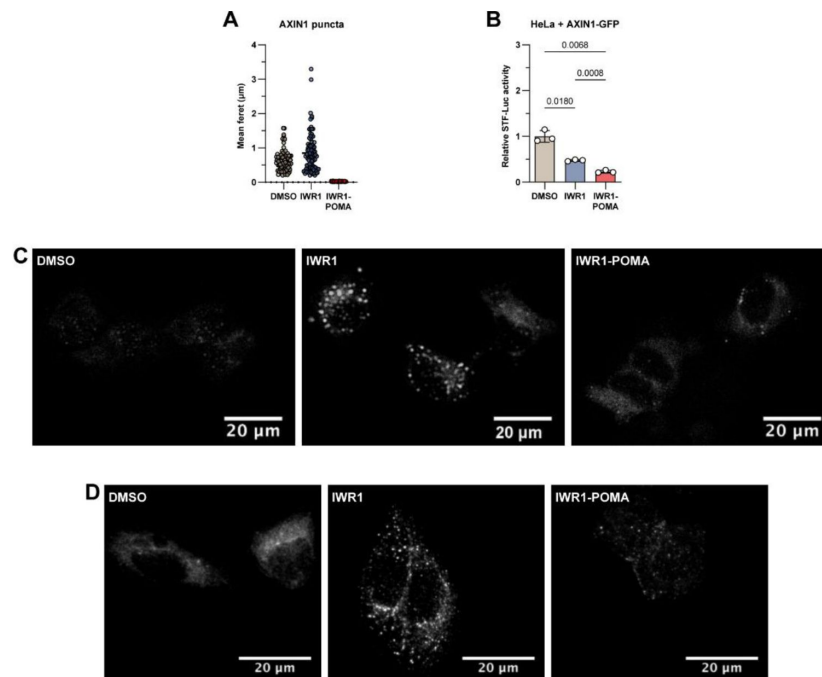


Figure S6.

TNKS*i* induced the formation of AXIN puncta

(A) Puncta count of samples corresponding to [Figure 4C](#). (B) HeLa cells transfected with AXIN1-GFP and STF plasmids and then treated with DMSO, IWR1 (3 μM) or IWR1-POMA (3 μM). IWR1-POMA suppressed WNT/β-catenin signaling significantly better than IWR1, indicating that AXIN puncta formation is not required for the DC to promote β-catenin degradation. The data is presented as mean ± SEM with p-values calculated by two-tailed unpaired t-test. (C) HeLa cells transfected with GFP-AXIN1 followed by treating with DMSO, IWR1 (3 μM) or IWR1-POMA (3 μM). Together with [Figure 4C](#), this experiment shows that the position of GFP tag does not affect puncta formation. (D) HeLa cells transfected with GFP-AXIN2 followed by treating with DMSO, IWR1 (3 μM) or IWR1-POMA (3 μM). This experiment shows that AXIN2 also forms puncta upon IWR1 treatment.

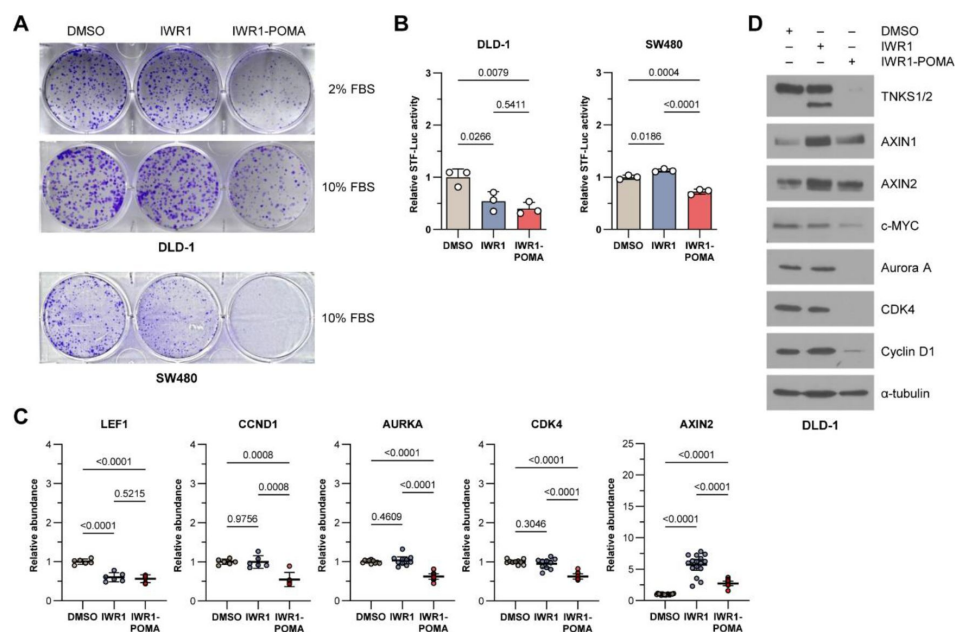


Figure S7.

IWR1-POMA suppressed CRC growth through WNT inhibition

(A) IWR1-POMA (3 μ M) suppressed DLD-1 and SW480 colony formation. (B) IWR1-POMA (3 μ M) suppressed of WNT signaling more effectively than IWR1 (3 μ M) in DLD-1 and SW480 cells. The data is presented as mean $\pm$ SEM with p-values calculated by two-tailed unpaired t-test. (C) Peptide abundance of WNT targets, corresponding to **Figure 1E** and **1F**. IWR1-POMA (3 μ M) controlled several WNT targets not regulated by IWR1 (3 μ M) in DLD-1 cells. (D) Western blot analysis confirmed that c-MYC, Aurora A, CDK4 and cyclin D1 responded to IWR1-POMA (3 μ M) but not IWR1 (3 μ M) in DLD-1 cells.

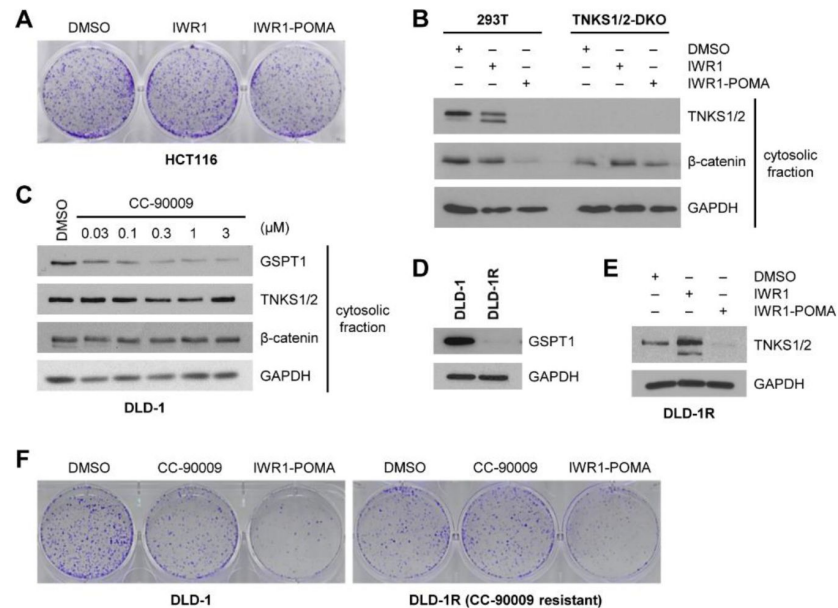


Figure S8.

IWR1-POMA suppressed CRC proliferation through on-target WNT inhibition

(A) HCT116 cells carrying a mutation in β -catenin that could not be processed by the DC were resistant to both IWR1 (3 μ M) and IWR1-POMA (3 μ M). (B) IWR1-POMA (3 μ M) reduced the cytosolic β -catenin level more effectively than IWR1 (3 μ M) in 293T cells. The level of cytosolic β -catenin in 293T-TNKS1/2-DKO cells that lack both TNKS1 and TNKS2 did not change with either drug treatment. (C) CC-90009 induced GSPT1 degradation in DLD-1 cells but had no effect on TNKS. GSPT2 was not detectable by Western blot, which is consistent with the reported GSPT levels determined by quantitative proteomic analysis in this cell line—102,567 ppb for GSPT1 and 2,495 ppb for GSPT2 (<https://www.ebi.ac.uk/gxa/experiments/E-PROT-18/Results>)⁵². (D) DLD-1R cells obtained from cultivating DLD-1 cells with CC-90009 have a dramatically reduced level of GSPT1 expression meanwhile maintaining normal TNKS expression. (E) IWR1 (3 μ M) promoted TNKS accumulation and IWR1-POMA (3 μ M) induced TNKS degradation in DLD-1R cells. (F) IWR1-POMA prevented colony formation of DLD-1R cells deficient in GSPT1/2.

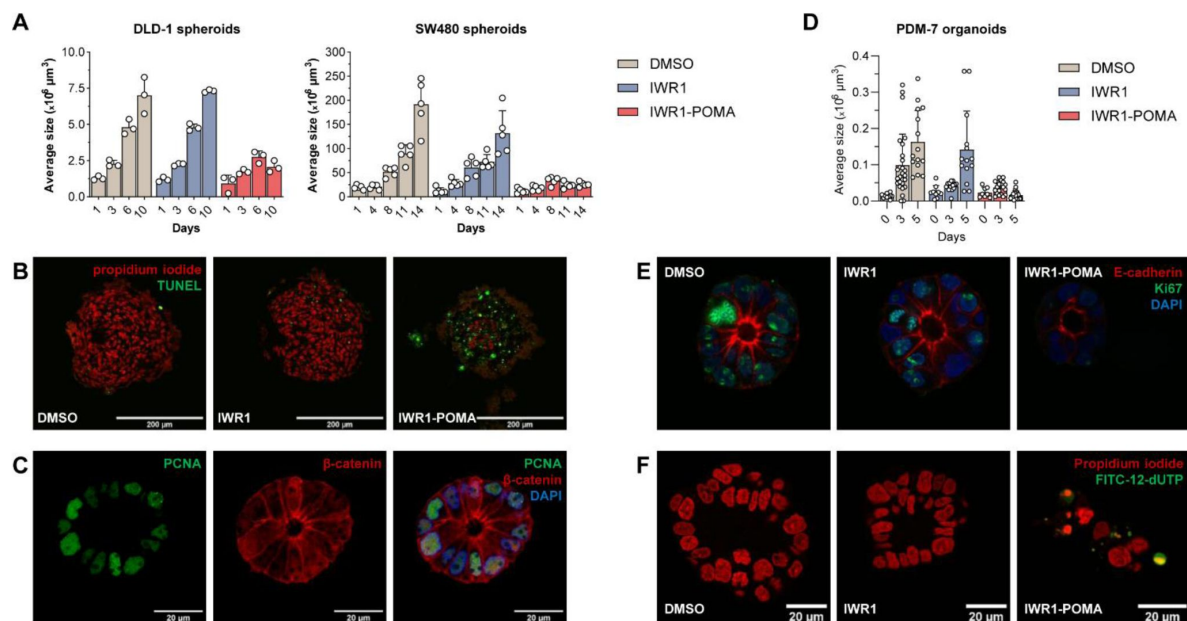


Figure S9.

IWR1-POMA demonstrated efficacy in CRC spheroid and primary organoid models

(A) The growth chart of DLD-1 and SW480 spheroids treated with DMSO, IWR1 (5 μM), or IWR1-POMA (5 μM). The sizes represent the apparent dimensions of the spheroids including the peripheral dead cells. (B) IWR1-POMA (5 μM) induced apoptosis in DLD-1 spheroids. (C) PDM-7 organoids grown from single cells preserved the heterogeneous nature of CRC tumors. (D) IWR1-POMA (1 μM) prevented the formation of PDM-7 organoids from single cells while IWR1 (1 μM) did not. (E and F) IWR1-POMA (1 μM) suppressed proliferation and induced apoptosis in PDM-7 organoids grown from single cells while IWR1 (1 μM) had little effect.

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Reviewer #1 (Public review):

Summary:

This manuscript reports the discovery and characterization of the first bifunctional degrader of tankyrase. Notably, the tankyrase degrader exhibits stronger β -catenin inhibition and tumor growth suppression compared to conventional tankyrase inhibitors. Mechanistically, while tankyrase inhibitors stabilize tankyrase and promote Axin puncta formation - thereby impairing β -catenin degradation - the degrader avoids this effect, resulting in deeper suppression of β -catenin signaling. These findings suggest that targeted degradation of tankyrase offers a novel therapeutic strategy for β -catenin-driven cancers. Overall, this is a compelling study with significant translational potential.

Strengths:

- (1) The manuscript presents a rigorous and well-executed study on a timely and impactful topic.
- (2) The biochemical and cellular characterization of the tankyrase degrader is thorough, and the comparative analysis with tankyrase inhibitors is insightful.
- (3) The finding that tankyrase stabilization by inhibitors may interfere with Axin function is novel and significant. It aligns with earlier observations (e.g., Huang 2009) that transient tankyrase overexpression can stabilize β -catenin independently of PAR domain activity.
- (4) The use of TNKS1/2 knockout cells expressing catalytically inactive tankyrase to demonstrate β -catenin inhibitory activity of the tankyrase degrader is elegant.
- (5) The finding that the tankyrase degrader has superior anti-proliferative effects in colorectal cancer models has important therapeutic implications.

Weaknesses:

(1) A key caveat is that the identified tankyrase degrader also targets GSPT1 for degradation. This raises the possibility that GSPT1 degradation may contribute to the observed β -catenin and tumor growth inhibition.

(2) The authors address this concern reasonably by showing that DLD1 cells resistant to GSPT1 degradation remain sensitive to the tankyrase degrader.

(3) To further strengthen this point, the authors might consider generating TNKS1/2 double knockout cells (e.g., in DLD1 or SW480 backgrounds) and demonstrating that the degrader loses its growth-inhibitory effect in these models. However, given the technical challenges of creating double knockouts in cancer cell lines, such experiments could be considered optional.

<https://doi.org/10.7554/eLife.109052.1.sa3>

Reviewer #2 (Public review):

Summary:

The ADP-ribosyltransferase tankyrase controls many biological processes, many of which are relevant to human disease. This includes Wnt/ β -catenin signalling, which is dysregulated in many cancers, most notably colorectal cancer. Tankyrase is a positive regulator of Wnt/ β -catenin signalling in that it counters the activity of the β -catenin destruction complex (DC). Catalytic inhibition of tankyrase not only blocks PAR-dependent ubiquitylation and degradation of AXIN1/2, the central scaffolding protein in the DC, but also tankyrase itself. As a result, blocking tankyrase gives rise to tankyrase accumulation, which may accentuate its non-catalytic functions, which have been proposed to drive Wnt/ β -catenin signalling. Most tankyrase catalytic inhibitors have shown limited efficacy and substantial toxicity in vivo. By developing tankyrase-directed PROTACs, the authors aim to block both catalytic and non-catalytic functions of tankyrase, aspiring to achieve a more complete inhibition of Wnt/ β -catenin signalling. The successfully developed PROTAC, based on the existing catalytic inhibitor IWR1, IWR1-POMA, induces the degradation of both TNKS and TNKS2, blocks β -catenin-dependent transcription without stabilising the DC in puncta/degradasomes, and inhibits cancer cell growth in vitro. Mechanistically, this points to a scaffolding role of tankyrase in the DC, at least under conditions of tankyrase catalytic inhibition, in line with previous proposals.

Strengths:

The study clearly illustrates the incentive for developing a tankyrase degrader, namely, to abolish both catalytic and non-catalytic functions of tankyrase. By and large, the study achieves these ambitions, and the findings support the main conclusions, although the statement that a more complete inhibition of the pathway is achieved requires corroboration. The proteomics studies are powerful. IWR1-POMA constitutes a very useful tool to re-evaluate targeting of tankyrase in oncogenic Wnt/ β -catenin signalling. The paired compounds will benefit investigations of tankyrase scaffolding functions across many different biological systems controlled by tankyrase. The findings are exciting.

Weaknesses:

Although the results are promising and mostly compelling, the claim that the PROTACs provide "a deeper suppression of the WNT/ β -catenin pathway activity" requires further corroboration, particularly at endogenous tankyrase levels.

There are also some other points that, if considered, would further improve the manuscript, as detailed below.

(1) Abstract and line 62: Many catalytic tankyrase inhibitors tend to display toxicity, which is likely on-target (e.g., 10.1177/0192623315621192; 10.1158/0008-5472). This constitutes the main limiting factor for these compounds. An incomplete inhibition of Wnt/beta-catenin signalling may contribute to the challenges, but this does not appear to be the dominant problem. A more prominent introduction to this important challenge is probably expected by the field.

(2) The authors do a good job in setting the scene for the need for tankyrase degraders. Their observations relating to the formation of puncta (degradasomes) being tankyrase-dependent are compatible with a previous study by Martino-Echarri et al. 2016 (10.1371/journal.pone.0150484): simultaneous silencing of TNKS and TNKS2 by RNAi abolishes degradasome formation. The paper is cited as reference 17, but only in passing, and deserves more prominence. (It includes an entire paragraph titled "Expression of tankyrases 1 and 2 is required for TNKSi-induced formation of axin puncta").

(3) Moreover, the scaffolding concept has been discussed comprehensively in other studies: 10.1111/bph.14038 and more recently 10.1042/BCJ20230230. There are also a few studies that focus on targeting the ankyrin repeat clusters of tankyrase to disengage substrates (10.1038/s41598-020-69229-y; 10.1038/s41598-019-55240-5) that illustrate the concept of blocking the scaffolding function. In that sense, the hypotheses are mature, and it is interesting to see some of them supported in this study. The authors could improve how they set their work into the context of these other efforts and proposals.

(4) In several places in the manuscript, the DC is referred to as "biomolecular condensate", at times even as a "classic example", implying that it operates through phase separation. This has not been demonstrated. In fact, super-resolution microscopy indicates that the puncta are not droplet-like (10.7554/eLife.08022), which would argue against the condensate hypothesis.

(5) It is beautiful to be able to use IWR1 and IWR1-POMA at identical concentrations for direct comparisons. However, this requires the two compounds to bind to tankyrase similarly well and reach the target to a comparable extent. How sure are authors that target engagement is comparable? Has this been evaluated?

(6) Figure 1F: It is not immediately apparent how IWR1-POMA shows more complete containment of Wnt/beta-catenin signalling. Most Wnt/beta-catenin targets lie close to the perfect diagonal, so I do not see how the statement "that IWR1-POMA controlled WNT/ β -catenin signaling more effectively than IWR1" (in the legend of Figure 1F) is supported. Minimally, an expanded explanation would benefit the reader. Providing the colour-coding legend directly in the figure would help improve clarity. Also, the panel is very small and may benefit from a different presentation in the figure.

(7) Figure 2: The conclusion of a "deeper suppression" of signalling relies on overexpression of tankyrase in an otherwise tankyrase-null background. Have the authors attempted to measure reporter activity or endogenous gene expression without tankyrase overexpression, in Wnt3a-stimulated cells (in the context of a normal Wnt/beta-catenin pathway) or CRC cells at the basal level? Non-catalytic activity in a similar assay has previously been observed upon tankyrase overexpression (10.1016/j.molcel.2016.06.019). Whether or not there is a substantial scaffolding effect at endogenous tankyrase levels after tankyrase inhibition remains unconfirmed, and the PROTAC is a valuable tool to address this important question. The findings presented in Figure S7C and D go some way towards answering this question - these data could be presented more prominently, and similar assays could be performed in other cell systems.

(8) Line 237/238: "TNKS accumulation negatively impacts the catalytic activity of the DC (Figure 5D)" - the data do not show this. Beta-catenin levels are a surrogate readout for DC function (phosphorylation and ubiquitylation). Minimally, this requires rewording, with reference to beta-catenin levels.

(9) Line 303-304: Beta-catenin is thought to exchange at beta-catenin degradasomes; this is clear from previous FRAP assays and the observation that phospho-beta-catenin accumulates in degradasomes upon proteasome inhibition (10.1158/1541-7786.MCR-15-0125). However, degradasome size hasn't, to my knowledge, been related to activity. Can this be clarified, please?

(10) There are previous hypotheses/proposals that the sensitivity of CRC cells to tankyrase inhibition correlates with APC truncation or PIK3CA status (10.1158/1535-7163.MCT-16-0578; 10.1038/s41416-023-02484-8). Have the authors considered expanding their cell line panel (Figure S7) to sample a wider range of cell lines, including some that are wild-type with regard to APC or Wnt/beta-catenin signalling in general? This would be a valuable addition to the work. Quantitated colony formation data could be moved to the main body of the manuscript.

(11) The manuscript only mentions toxicity (i.e., therapeutic window) in the last sentence of the Discussion section. As this is THE main challenge with tankyrase inhibitors (as mentioned above), can the authors expand their discussion of this aspect? Is there an expectation that PROTACs may be less toxic?

(12) Figures 3, 4, 5A: For fluorescence microscopy experiments, can these be quantified, and can repeat data be included?

(13) Figure 4, S6: An additional channel illustrating the distribution of cells (e.g., nuclei, cytoskeleton, or membrane) would be helpful for orientation and context for the AXIN1 signal.

(14) How were cytosolic fractions of cells prepared to assess cytosolic beta-catenin levels? This detail is missing from the methods.

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Reviewer #3 (Public review):

In this manuscript, Wang et al employ a chemical biology approach to investigate the differences between the enzymatic and scaffolding roles of tankyrase during Wnt β -catenin signalling. It was previously established that, in addition to its enzymatic activity, tankyrase 1/2 also plays a scaffolding function within the destruction complex, a property conferred by SAM-domain-dependent polymerization (PMID: 27494558). It is also known that TNKS1/2 is an autoregulated protein and that its enzymatic inhibition leads to accumulation of total TNKS proteins and stabilization of Axin punctae (through the scaffolding function of TNKS1/2), leading to rigidification of the DC and decreased β -catenin turnover. The authors surmised that this could, in part, explain the limited efficacy of TNKS1/2 catalytic inhibition for the treatment of colorectal cancers. To test this hypothesis, they evaluated a series of PROTAC molecules promoting the degradation of TNKS1/2 to block both the catalytic and scaffolding activities. They show that IWR1-POMA (their most active molecule) promotes more efficient suppression of beta-catenin-mediated transcription and is more active in inhibiting colorectal cancer cell and CRC patient-derived organoids growth. Mechanistically, the authors used FRAP to demonstrate that catalytic inhibitors of TNKS led to a reduced dynamic assembly of the DC (rigidification), whereas IWR1-POMA did not affect the dynamics.

Overall, this is an interesting study describing the design and development of a PROTAC for TNKS1/2 that could have increased efficacy where catalytic inhibitors have displayed limited activity. Knowing the importance of the scaffolding role of TNKS1/2 within the destruction complex, targeting both the catalytic and scaffolding roles certainly makes sense. The manuscript contains convincing evidence of the different mechanisms of the PROTAC vs catalytic inhibitors. Some additional efforts to quantify several of the experiments and to indicate the reproducibility and statistical analysis would strengthen the manuscript. Ultimately, it would have been great to evaluate the *in vivo* efficacy of IWR1-POMA in an *in vivo* CRC assay (APCmin mice or using PDX models); however, I realize that this is likely beyond the scope of this manuscript.

I have some recommendations listed below for consideration by the authors to strengthen their study:

- (1) The title is slightly misleading, as it is already known that the scaffolding function of TNKS is important within the DC. The authors should consider incorporating the PROTAC targeting aspect in the title (e.g., PROTAC-mediated targeting of tankyrase leads to increased inhibition of betacat signaling and CRC growth inhibition).
- (2) The authors should comment in the manuscript on the bell-shaped curve obtained with treatment of cells with the PROTACs (Figure S2C). This likely indicates titrating of the targets within a bifunctional molecule with increasing concentration (and likely reveals the auto-inhibition conferred by the catalytic inhibition alone).
- (3) The authors comment that using G007-LK as warehead was unsuccessful, but they do not show data. Do the authors know why this was the case?
- (4) Throughout the manuscript, the authors need to do a better job at quantifying their results (i.e., the western blots and the IF). For example, the degradation of TNKS1/2 in Figure 1D is not overly convincing. Similarly, the IF data in Figure 3 needs to be quantified in some ways. Along the same lines, the effect of IWR1-POMA treatments on the proliferation of cells and organoids should be quantified using viability assays... There is also no indication of how many times these experiments were performed and whether the blots shown are representative experiments. The quantification should include all experiments.

<https://doi.org/10.7554/eLife.109052.1.sa1>

Author response:

Reviewer #1 (Public Review):

We thank the Reviewer for the favorable feedback. The major concern is the collateral degradation of GSPT1. As the Reviewer noted, IWR1-POMA was able to suppress colony formation in DLD-1 cells resistant to GSPT1/2 degrader, suggesting that TNKS but not GSPT degradation is responsible for growth inhibition.

We also appreciate that the Reviewer brought it to our attention an important early observation of the TNKS scaffolding effects. Cong reported in 2009 that overexpression of TNKS induced AXIN puncta formation in a SAM but not PARP domain-dependent manner (PMID 19759537). We will include this information in the revised manuscript.

Reviewer #2 (Public Review):

We thank the Reviewer for the encouraging and insightful comments. The major critique concerns whether TNKS degraders can suppress WNT/ β -catenin signaling more effectively than TNKS inhibitors at endogenous TNKS levels. Fig. 1D shows that IWR1-POMA reduced the level of cytosolic β -catenin more effectively than IWR1 in Wnt3A-stimulated HEK293 cells without protein overexpression, and Fig. S7B shows that IWR1-POMA reduced STF signals more effectively than IWR1 in DLD-1 and SW480 cells with endogenous TNKS expression. We will corroborate these findings with additional cell lines during the revision.

(1) We agree with the Reviewer that on-target toxicities pose challenges to the development of WNT inhibitors. For example, LGK974 that inhibits PORCN to prevent the secretion of all WNT proteins showed significant on-target toxicity in human (PMC10020809), and G007-LK that inhibits TNKS to block canonical WNT signaling selectively exhibited weak efficacy and dose-limiting toxicity at 5–30 mg/kg BID or 10–60 mg/kg QD in various mouse xenograft models (PMID: 23539443). Similarly, G-631, another TNKS inhibitor, also showed dose-limiting toxicity without significant efficacy at 25–100 mg/kg QD in mice (PMID: 26692561). However, G007-LK was well-tolerated at 200 mg/kg QD over 3 weeks in mice in another study (PMC5759193). Treating mice with G007-LK at 10 mg/kg QD over 6 months also improved glucose tolerance without notable toxicity (PMID 26631215). Importantly, constitutive silencing of both TNKS for 150 days in *APC*-null mice prevented tumorigenesis without damaging the intestines (PMC6774804). Furthermore, basroparib, a selective TNKS inhibitor, was well tolerated in a recent clinical trial (PMC12498271). We are therefore cautiously optimistic that TNKS degraders will have an improved therapeutic index compared with TNKS inhibitors.

(2) We agree with the Reviewer that Henderson's 2016 paper (PMC4773256) shed important light on the role of TNKS scaffolding in the DC. However, whereas this study demonstrated that knocking down both TNKS by siRNA prevented G007-LK to induce AXIN puncta, the function role of TNKS scaffolding in the DC remained unaddressed. We will include a more detailed description on Henderson's discovery.

(3) Indeed, Guettler demonstrated that TNKS scaffolding could promote WNT/ β -catenin signaling in 2016, which forms the basis of the current work. Meanwhile, whereas there have been efforts to target the SAM or ARC domain to address TNKS scaffolding, our approach of targeting TNKS for degradation is complementary. We will provide a more detailed discussion of these studies.

(4) Biomolecular condensates are membrane less cellular compartments formed by phase separation of biomolecules, regardless of the physical/material properties (PMID: 28935776 and PMC7434221). Super-resolution microscopy studies by Peifer and Stenmark (PMC4568445 and PMID 26124443) showed that AXIN, APC, TNKS, and β -catenin interacted with each other to assemble into membraneless complexes, wherein AXIN and APC formed filaments throughout the DC. Peifer has also summarized evidence that supports the condensate nature of the DC (PMC6386181). However, we acknowledge that testing the physical properties of reconstituted DC (PMC8403986) will provide a better understanding of the nature, for example liquid vs. gel, of these condensates.

(5) We will evaluate the ability of IWR1 and IWR1-POMA to engage TNKS.

(6) We will modify Fig. 1F to improve clarity and readability.

(7) Fig. S7B shows that IWR1-POMA suppressed WNT/ β -catenin signaling more effectively than IWR1 in *APC*-mut DLD-1 and SW480 CRC cells without TNKS overexpression. Similarly, Fig. S6B shows that IWR1-POMA provided a deeper suppression of STF signals in HeLa cells

transfected with AXIN1 and β -catenin while expressing endogenous TNKS. These results provide evidence that inhibitor-induced TNKS scaffolding plays a significant role at endogenous TNKS expression levels. Separately, we will reorganize the figures to better present Fig. 7C and D as suggested by the Reviewer.

(8) We will rephrase "TNKS accumulation negatively impacts the catalytic activity of the DC".

(9) We apologize for confusing β -catenin phosphorylation with β -catenin abundance. Here, we refer the catalytic activity of the DC to as the ability of the DC to promote β -catenin degradation rather than the kinetics of β -catenin phosphorylation and ubiquitination. It is commonly observed that AXIN stabilization by TNKS inhibitors increases the DC size and reduces the β -catenin levels. Peifer has also noted that APC can increase the size and the "effective activity" of the DC (PMC5912785 and PMC4568445). As such, the induction of AXIN puncta by TNKS inhibitors is frequently used as an indicator of WNT/ β -catenin pathway inhibition. However, because the DC only primes β -catenin but does not catalyze its degradation, we will revise our manuscript to improve accuracy and clarity.

(10) We will examine the effects of IWR1 and IWR1-POMA in additional cell lines, quantify the colony formation data, and reorganize the figures.

(11) As discussed above, evidence for on-target toxicity of WNT/ β -catenin inhibition is mixed. Yet, the observation of no dose-limiting toxicity for basroparib at doses up to 360 mg QD in human (PMC12498271) is encouraging. PROTAC works by catalyzing target degradation, which is different from traditional catalytic inhibitors that require continuous target occupancy at a high level. Because IWR1-POMA has a durable effect on TNKS, we expect that a fully optimized TNKS degrader may allow less frequent dosing than basroparib and consequently an even more favorable therapeutic window.

(12/13) We will include quantification data, replicate information, and nuclei staining or cell outlines for the fluorescence microscopy experiments.

(14) Cytosolic fractions of cells were prepared using a commercial cytoplasmic extraction kit following manufacturer's instructions. We will include detailed information in the revised manuscript.

Reviewer #3 (Public Review):

We thank the Reviewer for the helpful suggestions.

(1) We will modify the title to include the PROTAC aspect.

(2) As the Reviewer suggested, the bell-shaped dose response of the PROTAC originated from the formation of saturated binary complexes. At high PROTAC concentrations, binding of TNKS and CRBN/VHL by separate PROTAC molecules impedes the formation of productive ternary complexes, which results in reduced degradation efficacy and consequently the hook effect.

(3) The structure-activity relationship of PROTACs is often unpredictable, as both the kinetics and thermodynamics of the target and E3 ligase binding play crucial roles. The lack of translation in degradation efficacy from IWR1 to G007-LK derived PROTACs may originate from differences in the binding kinetics or subtle changes in the orientation of the linker exit vector. We will include data on G007-LK in the revised manuscript.

(4) We will quantify the Western blots, immunofluorescence images, colony formation data, and the replicate information.

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